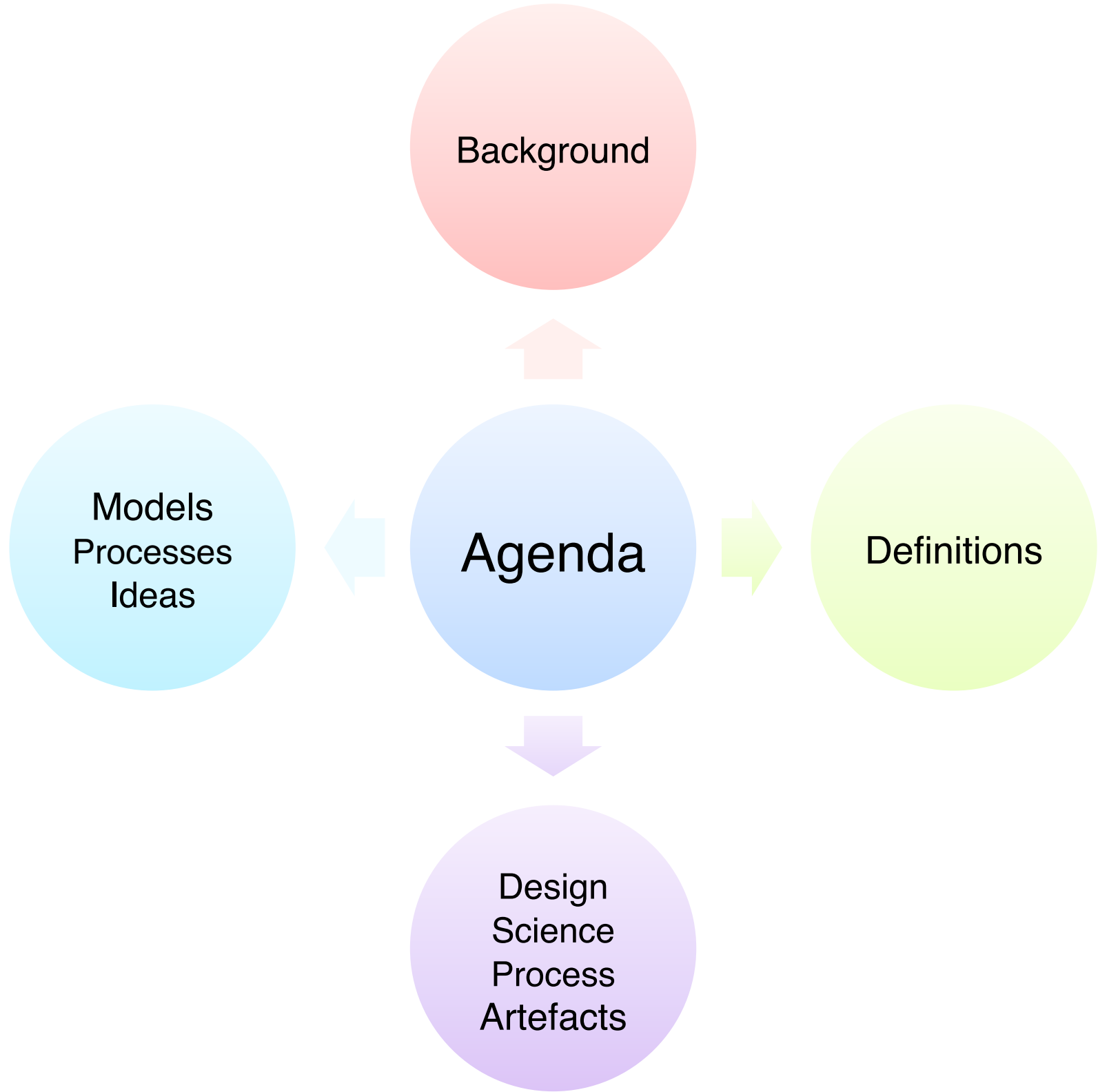
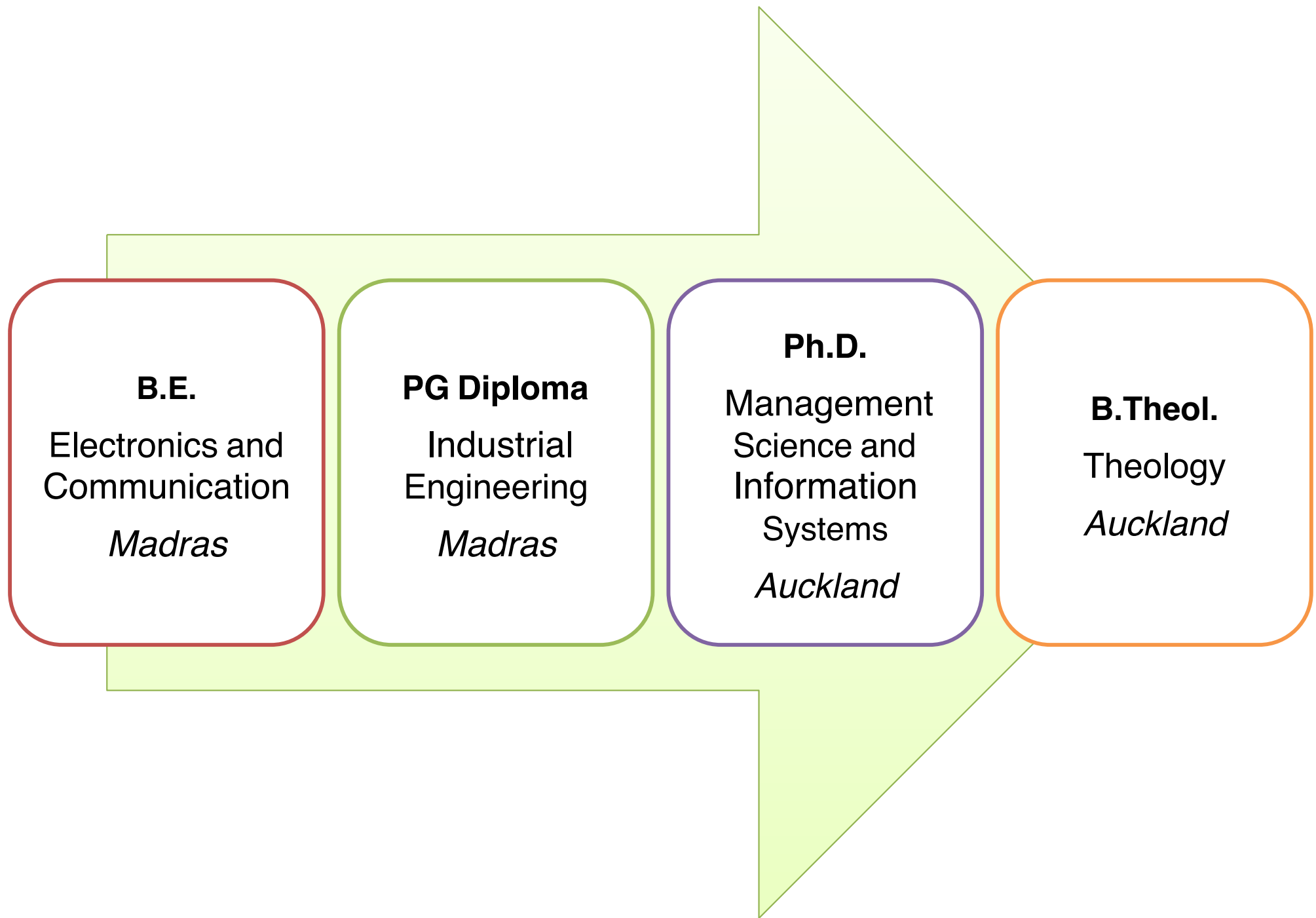
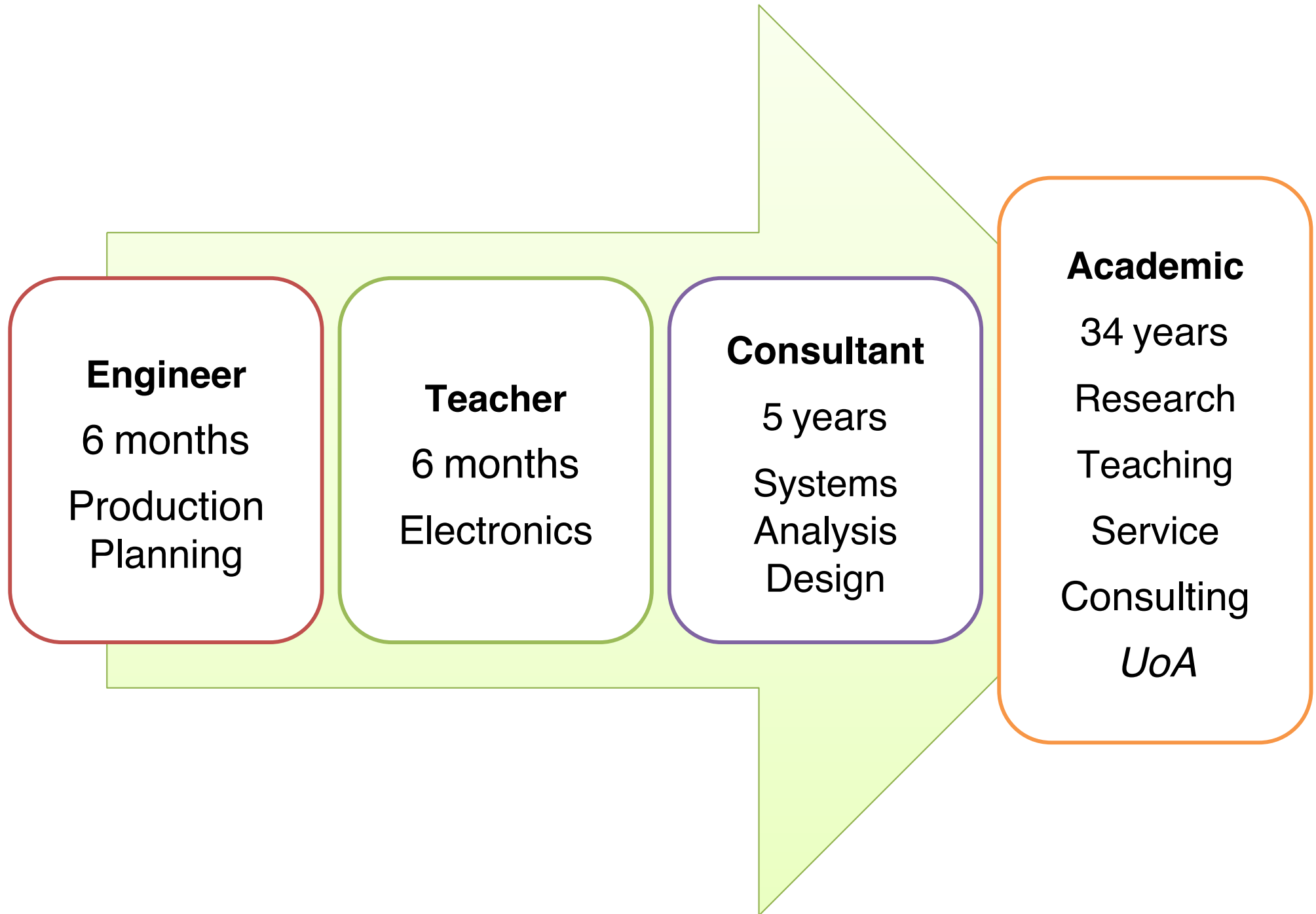


Design Science Research

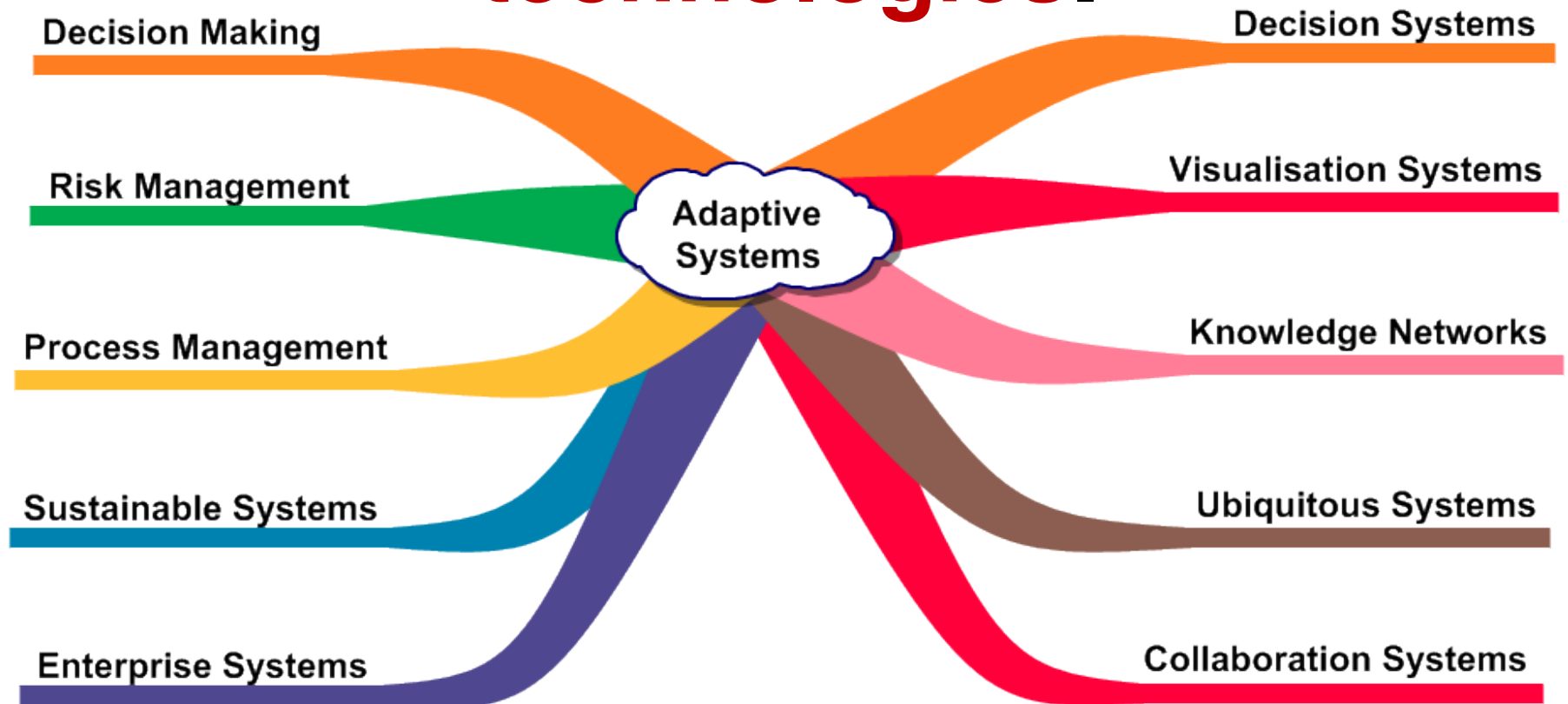








I am interested in the design and implementation of **adaptive sustainable systems** – interweaving **strategy**, **people**, **processes**, and **technologies**.



Visions

Strategies

Tactics

Processes

Roadmaps

Implementations

Models

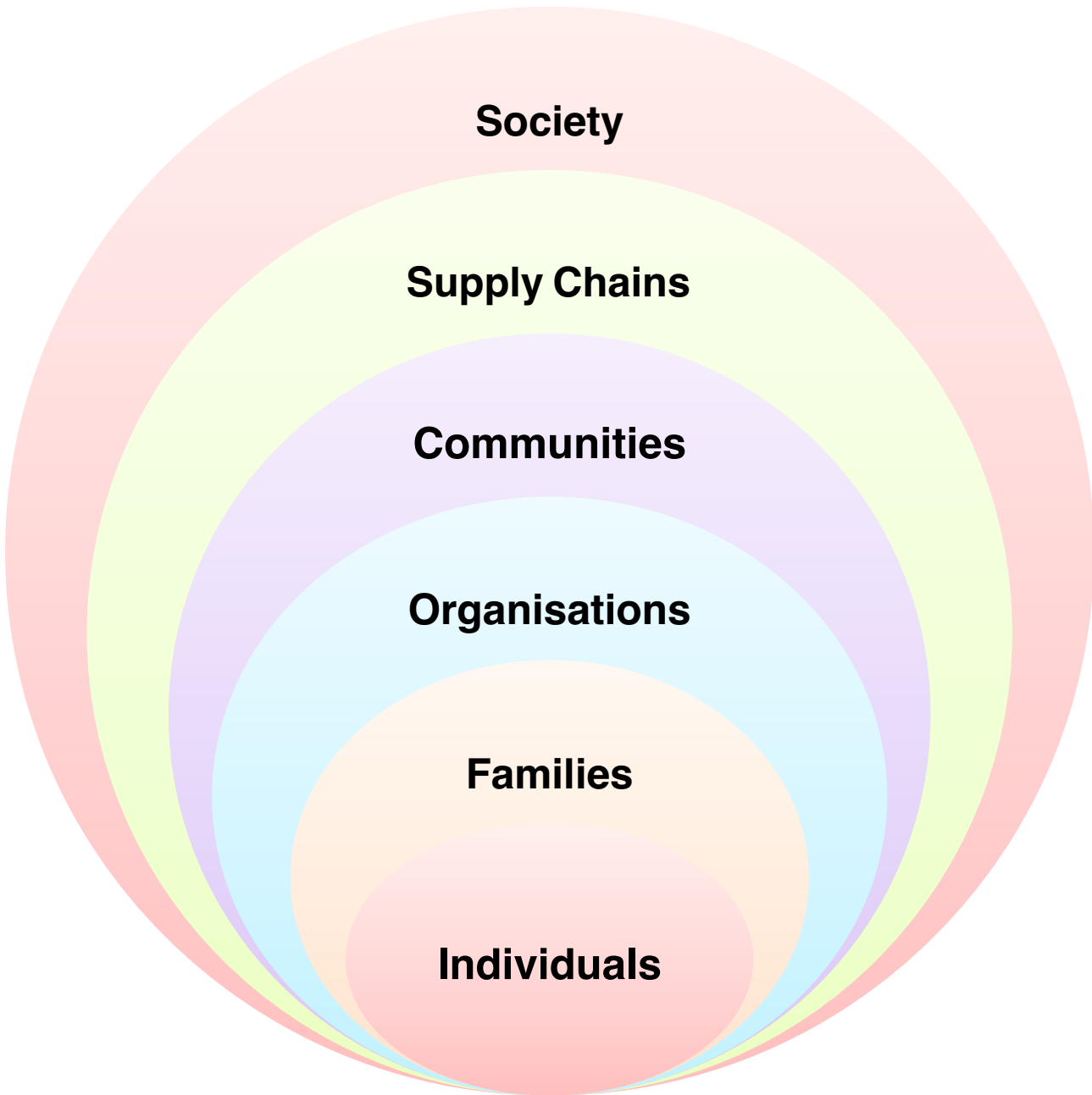
Paradigms

Systems

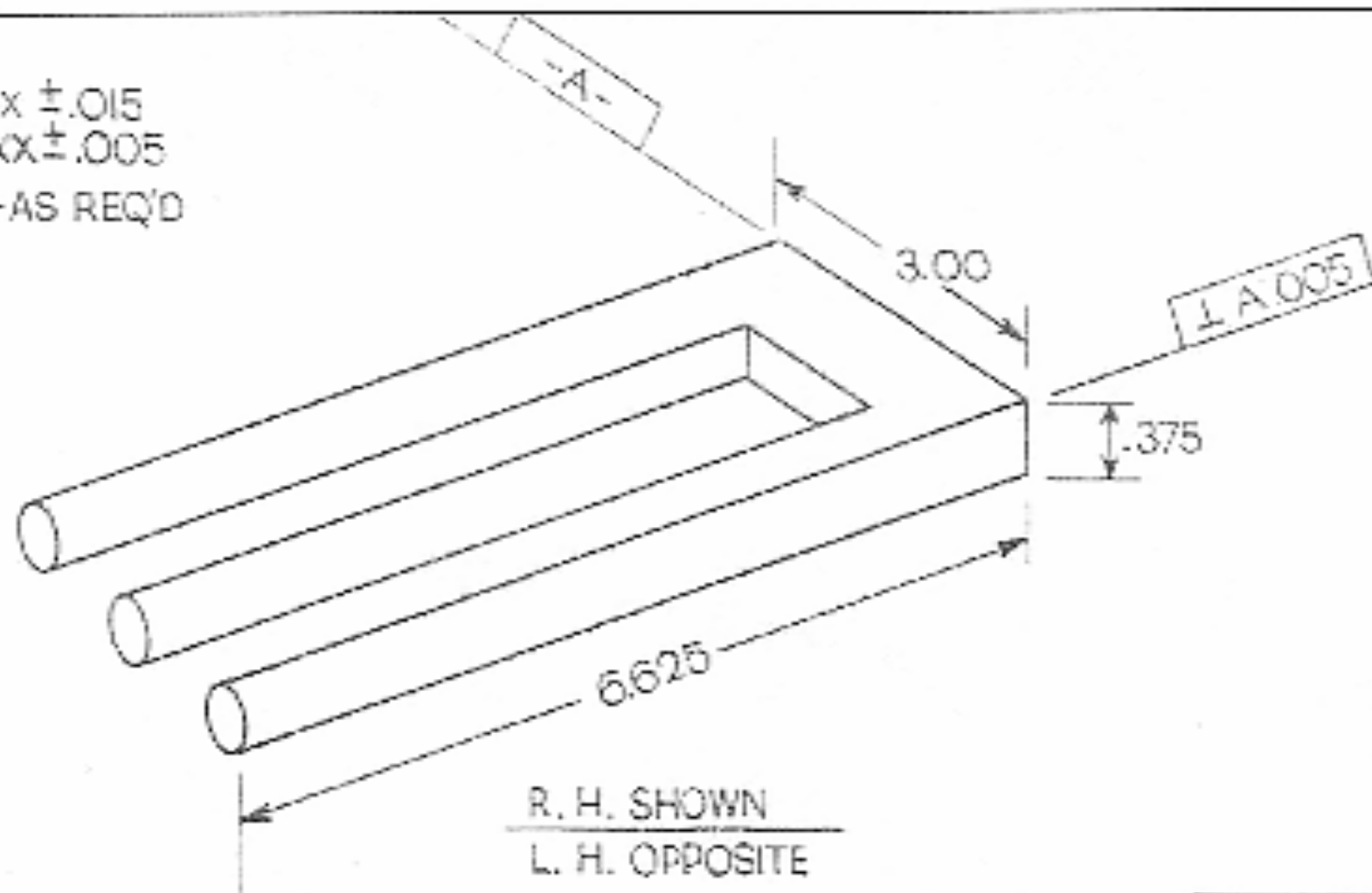
Applications

Architectures

Frameworks



TOLERANCES XX $\pm .015$
 XXX $\pm .005$
 BEND RADIUS - AS REQ'D



ASSY. NO. 41

IMC MAGNETICS CORP.

MATL. - ALUM ALLOY 6061-T6 OR 6062-T6

PER UNIT 16-TR REQ'D

DRAWN <i>WOK</i>	SCALE $\frac{1}{2}$	REVISIONS	DATE <i>1-10-4</i>
CHECKED <i>WOK</i>			
APPROVED <i>WOK</i>			
DATE <i>1-10-4</i>			
TITLE BRACE HORIZONTAL			No. IMC 1736-19

Definitions

What is
DESIGN?

Design is both

- a **process** (set of activities) and
- a **product** (artifact)

Design is both a **verb** and a **noun**

Design describes

- the world as **acted** upon (processes)
- the world as **sensed** (artifacts).

What is
DESIGN
SCIENCE?

Design-science seeks to
extend the boundaries of
human & organizational
capabilities by creating
new and **innovative**
artifacts.

What are

DESIGN SCIENCE

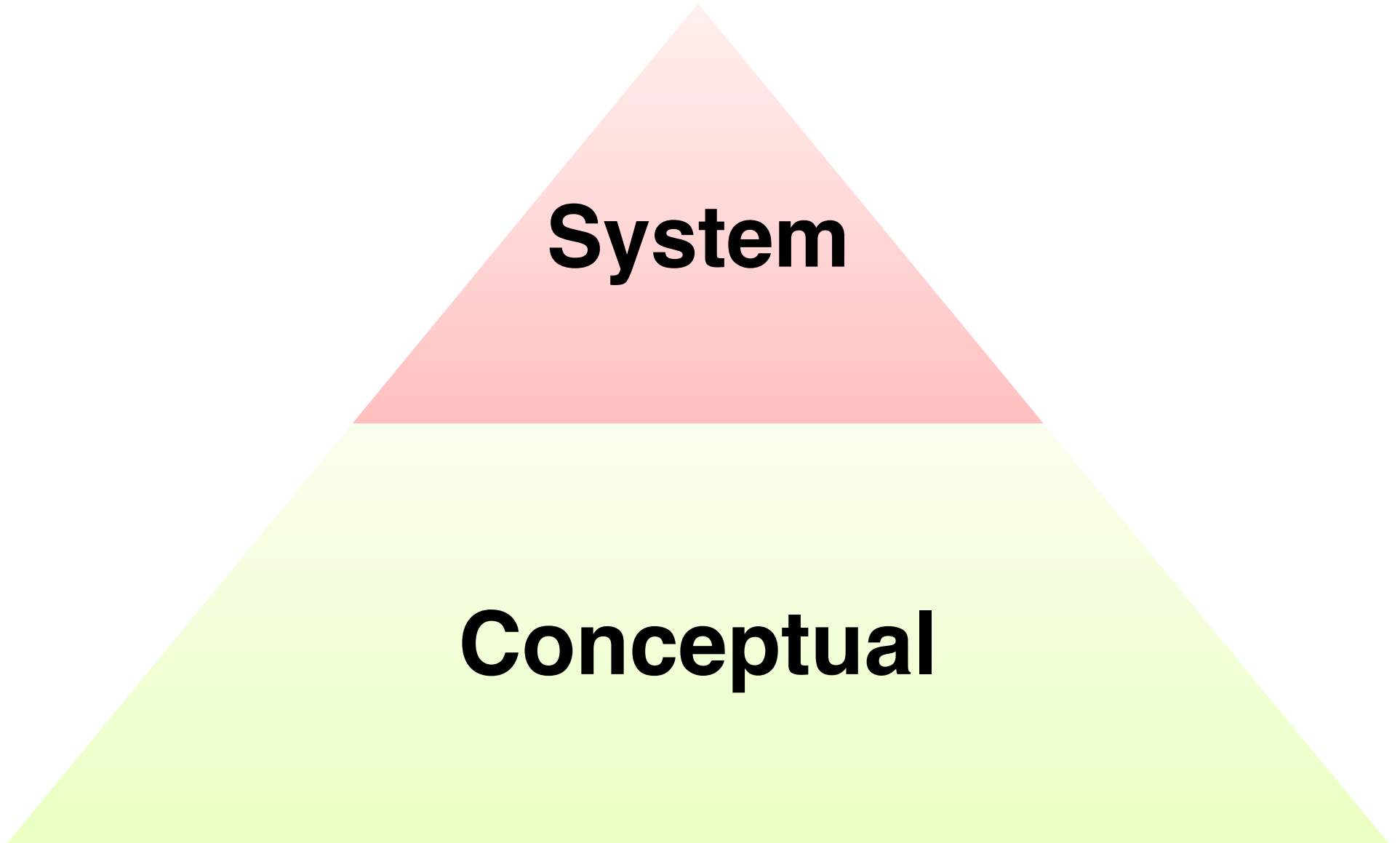
RESEARCH

ARTEFACTS?

Design Science research artifacts
are **INNOVATIONS** that **DEFINE**
the **concepts**, **models**, **processes**,
frameworks, **architectures**, and
systems through which the analysis,
design, implementation, and use of
information systems can be
effectively and efficiently
accomplished.

Adapted from Hevner, 2004; Denning 1997; Tsichritzis 1998

Broad Categories of Design Science Artefacts

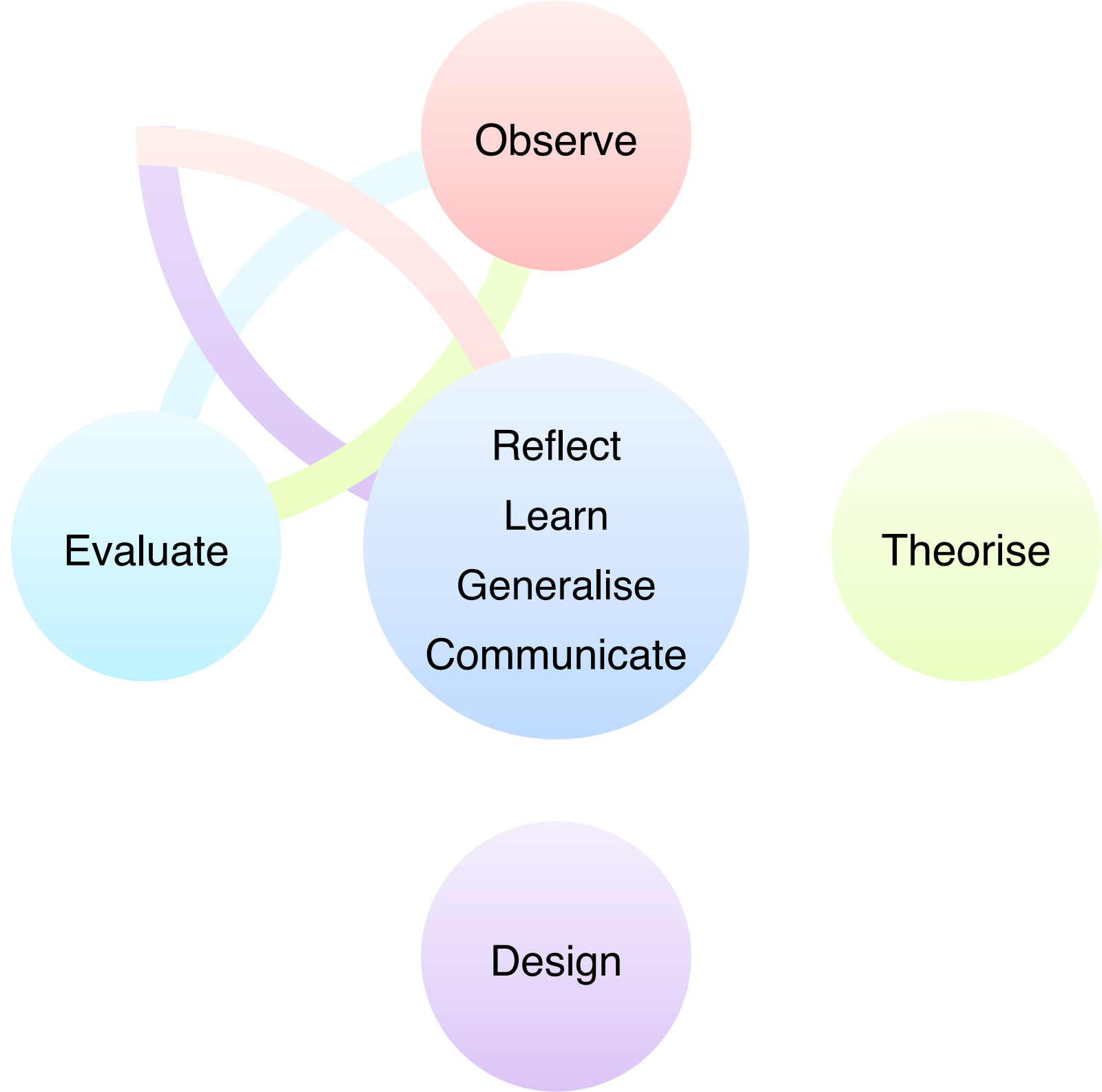


Design

Science

Process

Artefacts

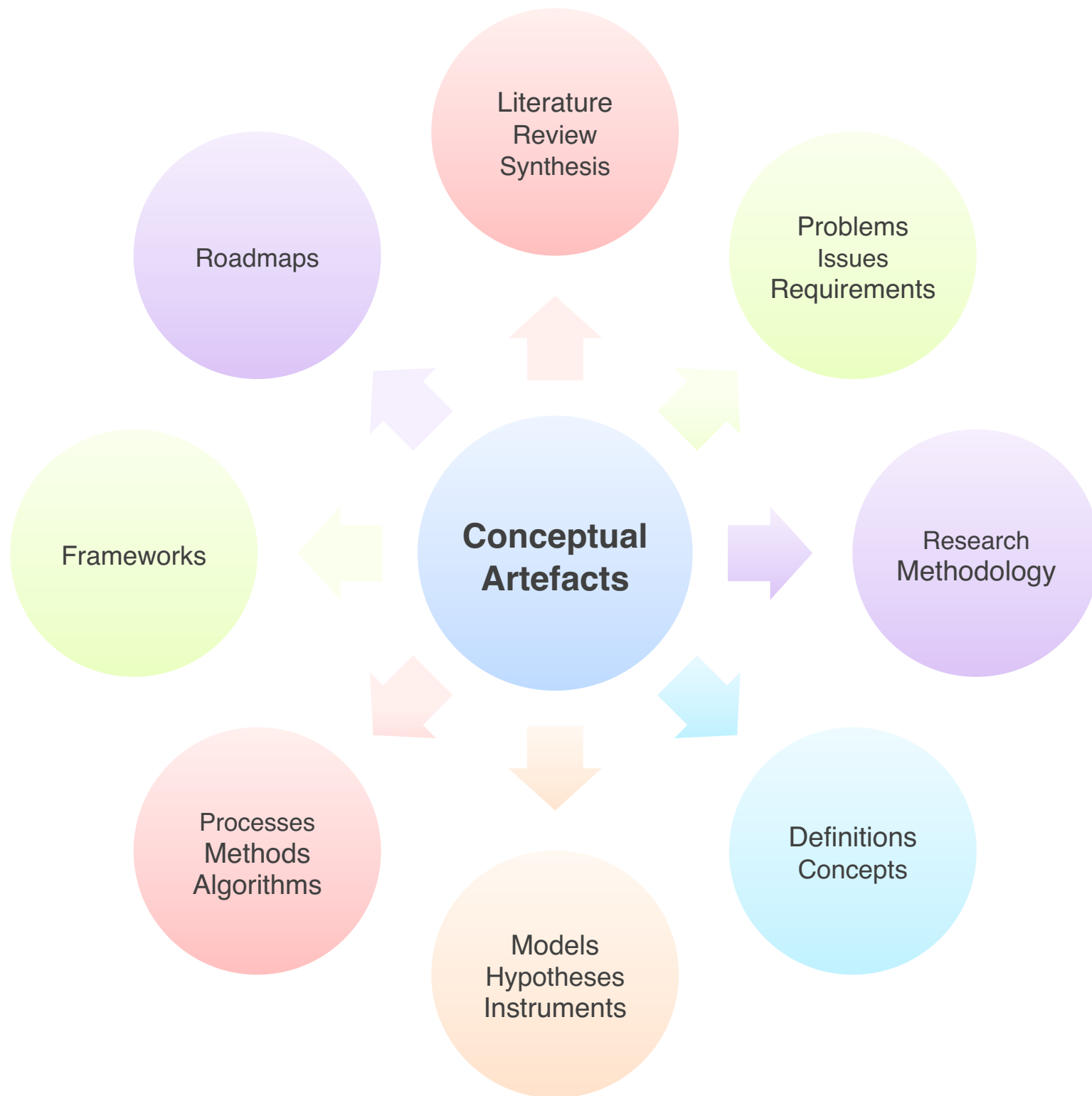


Observe

- **Systematic Literature Review – PRISMA, Rapid LR**
- Systematic Systems Review
- **Case studies/Field Studies**
- Surveys/Delphi Studies/Interviews
- **Experiments/Simulations**
- Experience
- **Research and state Users Needs and Problems**
- Identification of Research Problems, Issues, and Requirements

Theorise – Conceptual Artefacts

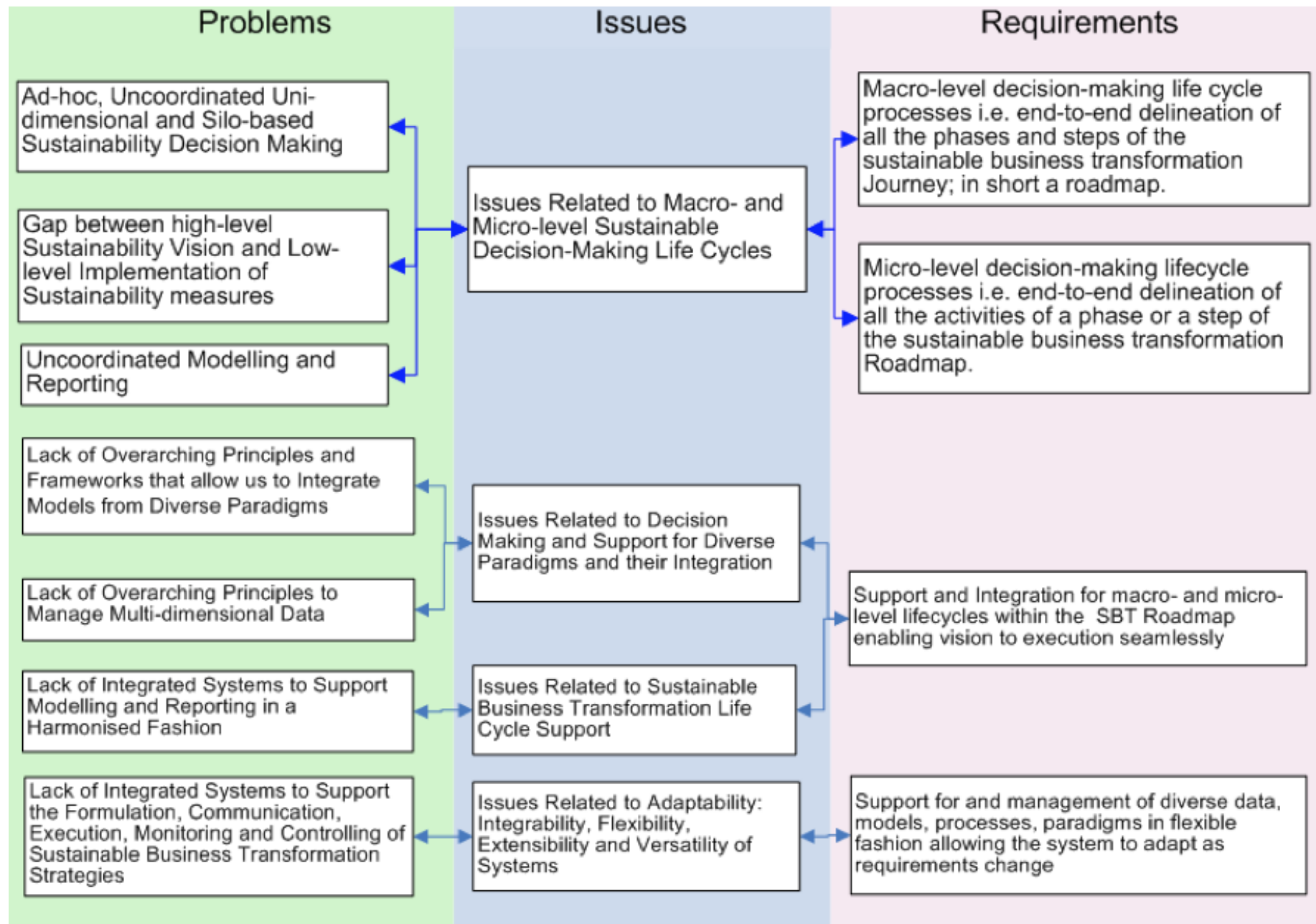
- Theorise
- Using theories from multiple disciplines -
INPUTS
- Building IS centric theories - OUTPUTS
- Contributing to IS and other disciplines
- Induction, Deduction, and Abduction
- Ideate/brainstorm
- https://is.theorizeit.org/wiki/Main_Page



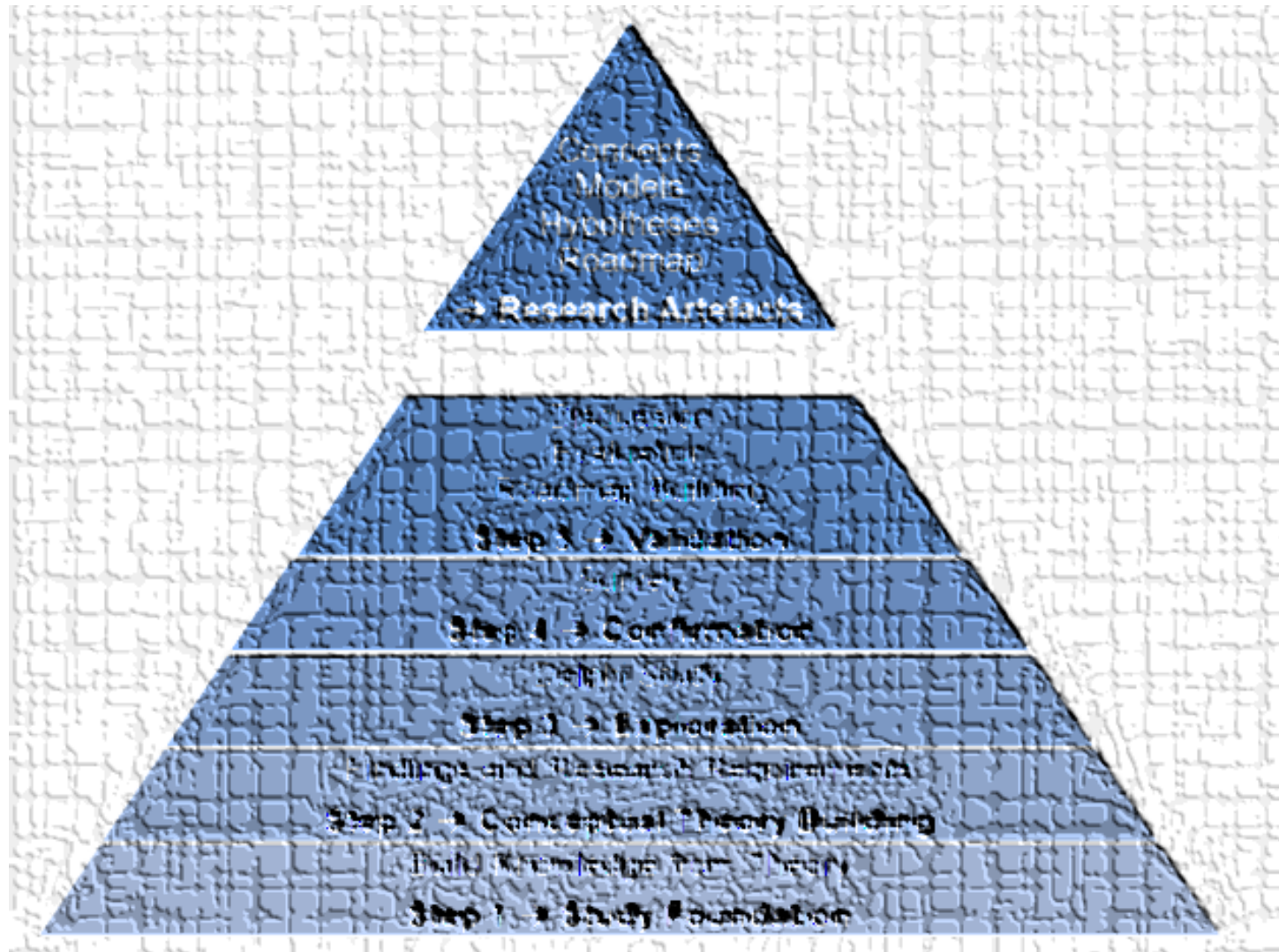
Literature Review Synthesis

Authors	Knowledge Sharing																								
	Knowledge	Management												Context											
		Knowledge Structure	Sharing	Knowledge Sharing Process				Knowledge Worker			Knowledge Management	Knowledge Sharing	Systems	Technological Environment	Socioeconomic environment		Organisational Environment			Organisational Structure	Context				
Communities	Networks	Groups				Knowledge codification	General		Individual	Experience		Motivation	Innovative		Traditional	Economical stability	Social Capital	Climate-Culture	Relationship		General	Risk Factor	Culture	Cognitive Factors	General
Polanyi (1958,1966)																									
DeSanctis and Gallupe (1987)																									
Ackoff (1989)																									
Neches et al. (1991)																									
Kogut and Zander (1992)																									
Nonaka (1994)																									
Blackler (1995)																									
Nelson and Coopridner (1996)																									
Spender (1996)																									
Davenport (1998);(Davenport and Prusak (2000)																									
Nonaka and Konno (1998)																									
Alavi and Leidner (1999)																									
Hendriks (1999)																									
Blackler et al. (2000)																									
DeLong andFahey(2000)																									
Dyer and Nobeoka (2000)																									
Nonaka and Toyama, et al. (2000)																									
Tiwana (2000)																									
Alavi and Leidner (2001)																									
Brown and Duguid (2001)																									
Ciborra and Andreu (2001)																									
Cross., Parker, et al. (2001)																									
Lesser and Storck (2001)																									
Walsham, (2001)																									
Grover & Davenport (2001)																									
Ruppel, C. P., S. J. Harrington, et al. (2001)																									
Cabrera & Cabrera (2002)																									
Davenport et al.(2002)																									
Goh (2002)																									
Hansen (2002)																									
Husted and Michailova (2002)																									
Tyndale (2002)																									
Bouthillier, and.Shearer (2002)																									
Ipe (2003)																									
Kalling & Styhre (2003)																									
Fernie, Green, et al. (2003).																									
Benbya, Passiante, et al. (2004).																									
Hooff and Ridder (2004)																									
Thompson and Walsham (2004)																									
Handzic (2004)																									
Sanchez (2004)																									
Bock, Zmud, Kim et al. (2005)																									
Hansen, et al. (2005)																									
Perez-Araos et al. (2007)																									
Chatti, Klamma, et al. (2007)																									
Ardichvili (2008)																									
Vithessonthi (2008)																									
Chow and Chan (2008)																									
Parise (2009)																									
Foss, Husted, et al. (2009)																									

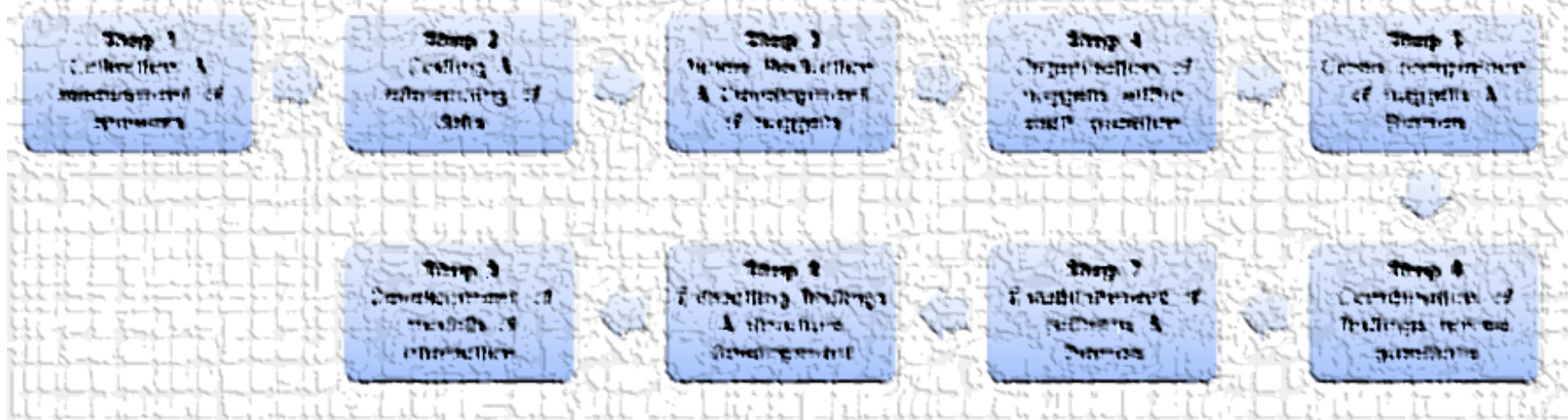
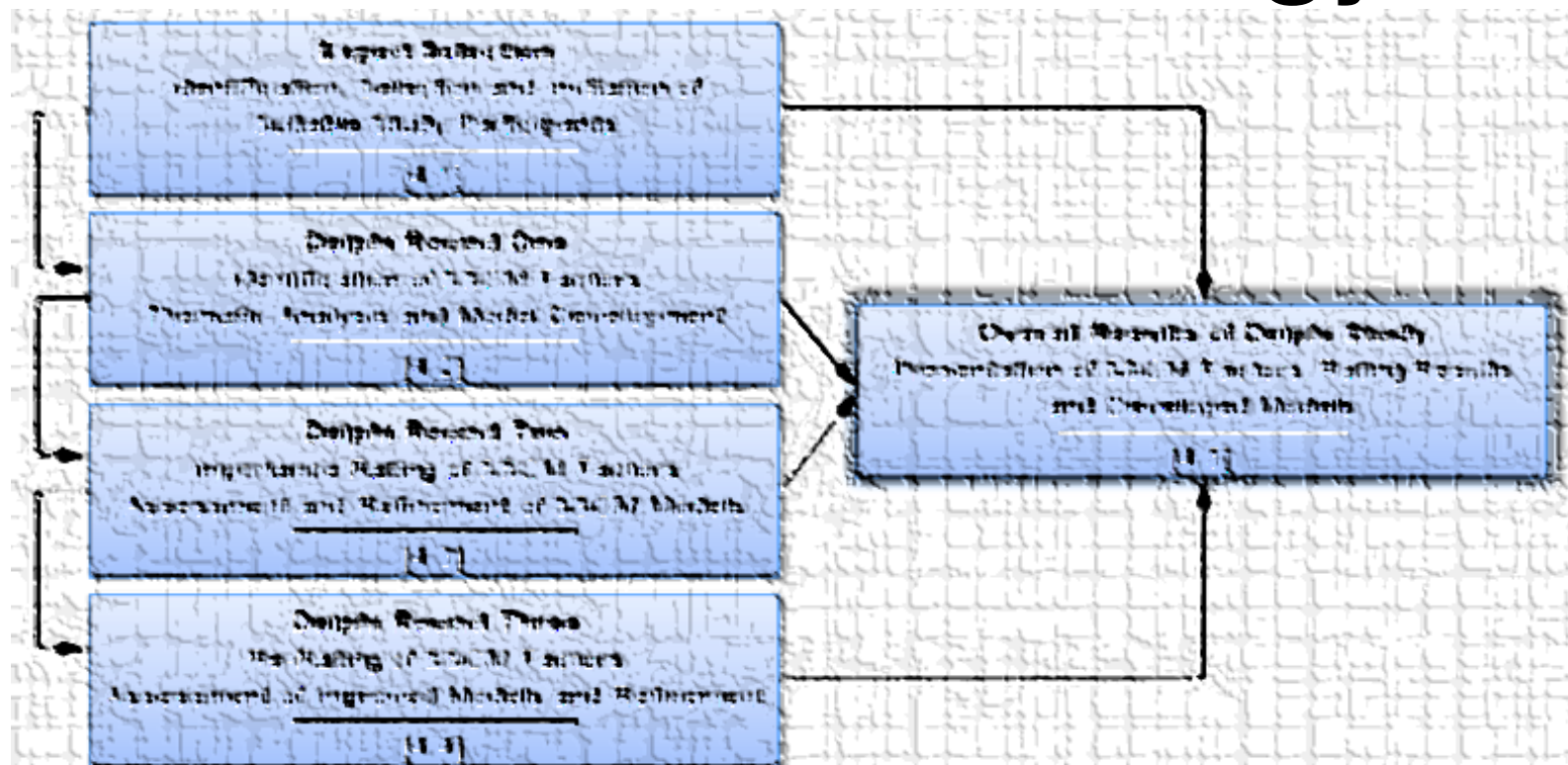
Problems Issues Requirements



Research Methodology



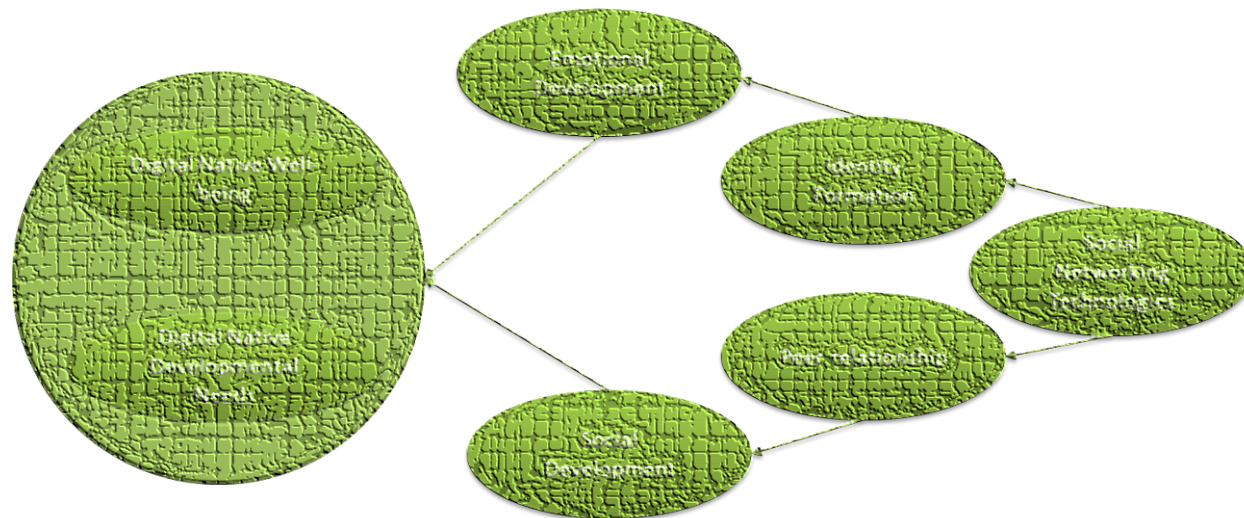
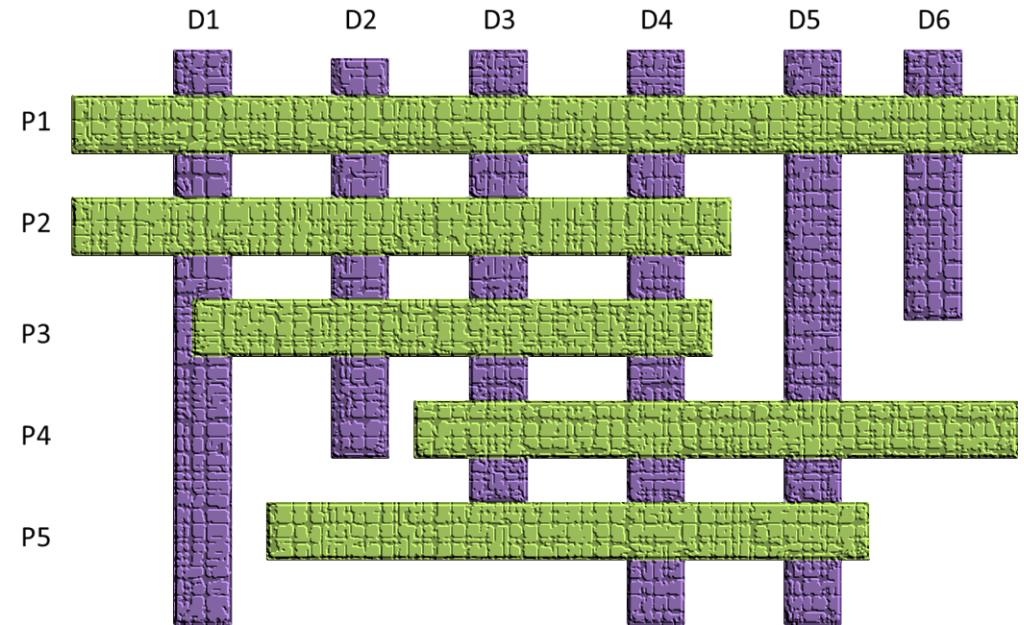
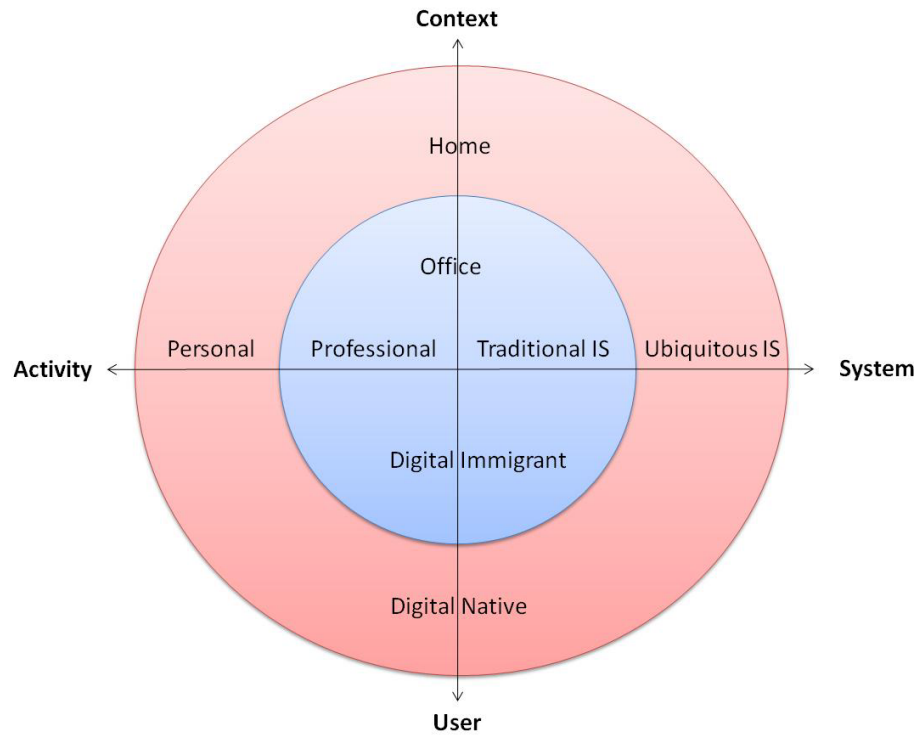
Research Methodology



Definitions and Concepts

- Visual Intelligence Density – BAI
- IOMS – Rad
- IADR - Rad
- Sustainable Social Shopping – Claris, Roman,..
- Sustainable Supply Chain Management – Hendrik
- Adaptive Sustainable Enterprise – Gabrielle, Jim,..

Models Hypotheses Instruments



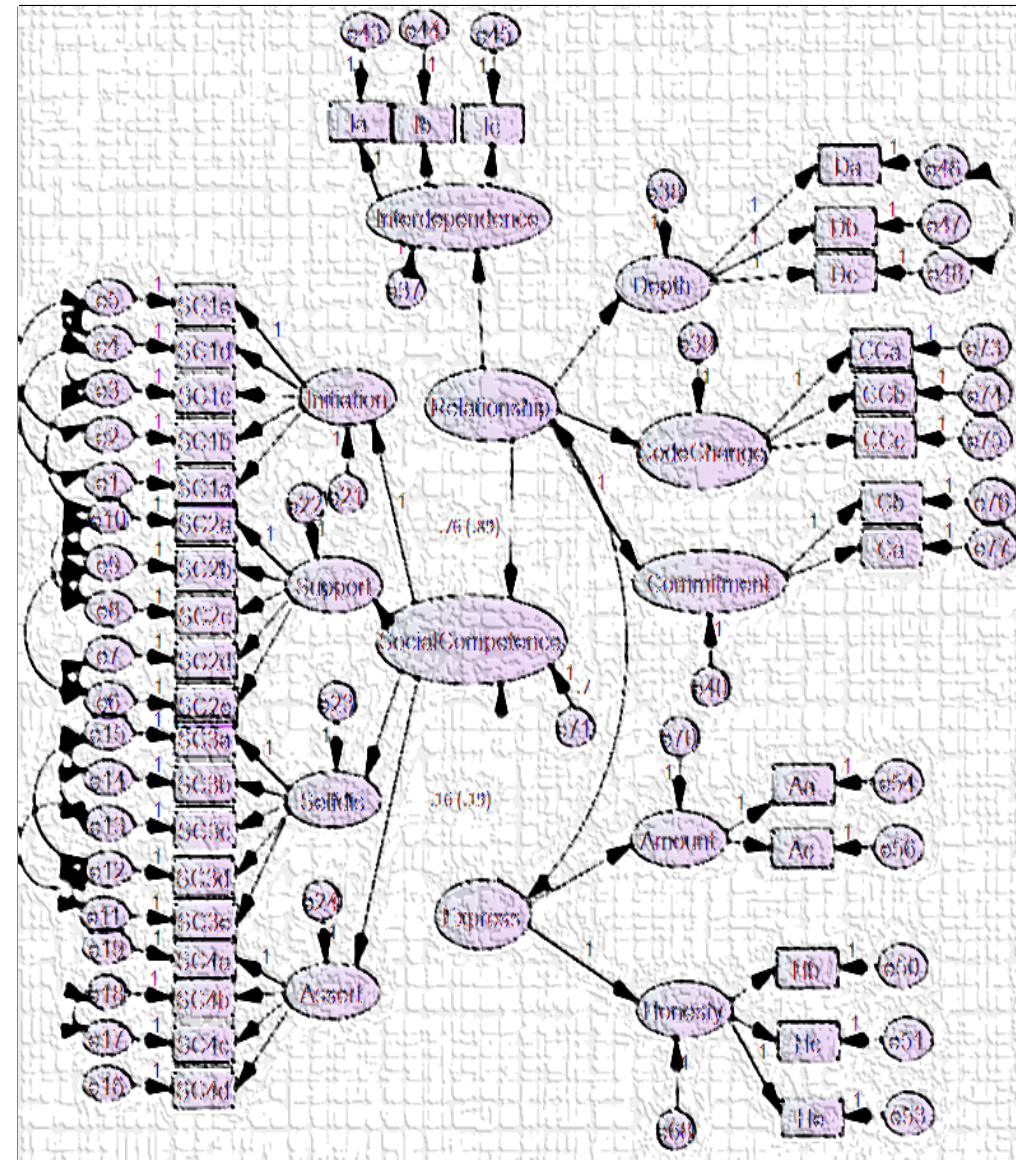
Models Hypotheses Instruments

H1: The higher the use of SNT the higher the social competence of Digital Natives

- H1a: The ability to form relationships through SNT leads to higher Social Competence in Digital Natives (Subrahmanyam and Greenfield 2008, Lee 2009)
- H1b: The ability to express oneself through SNT leads to higher Social competence in Digital Natives (Sullivan 2002; Davis 2010)

H2: The effect of SNT on Social Competence is moderated by Personality and Gender

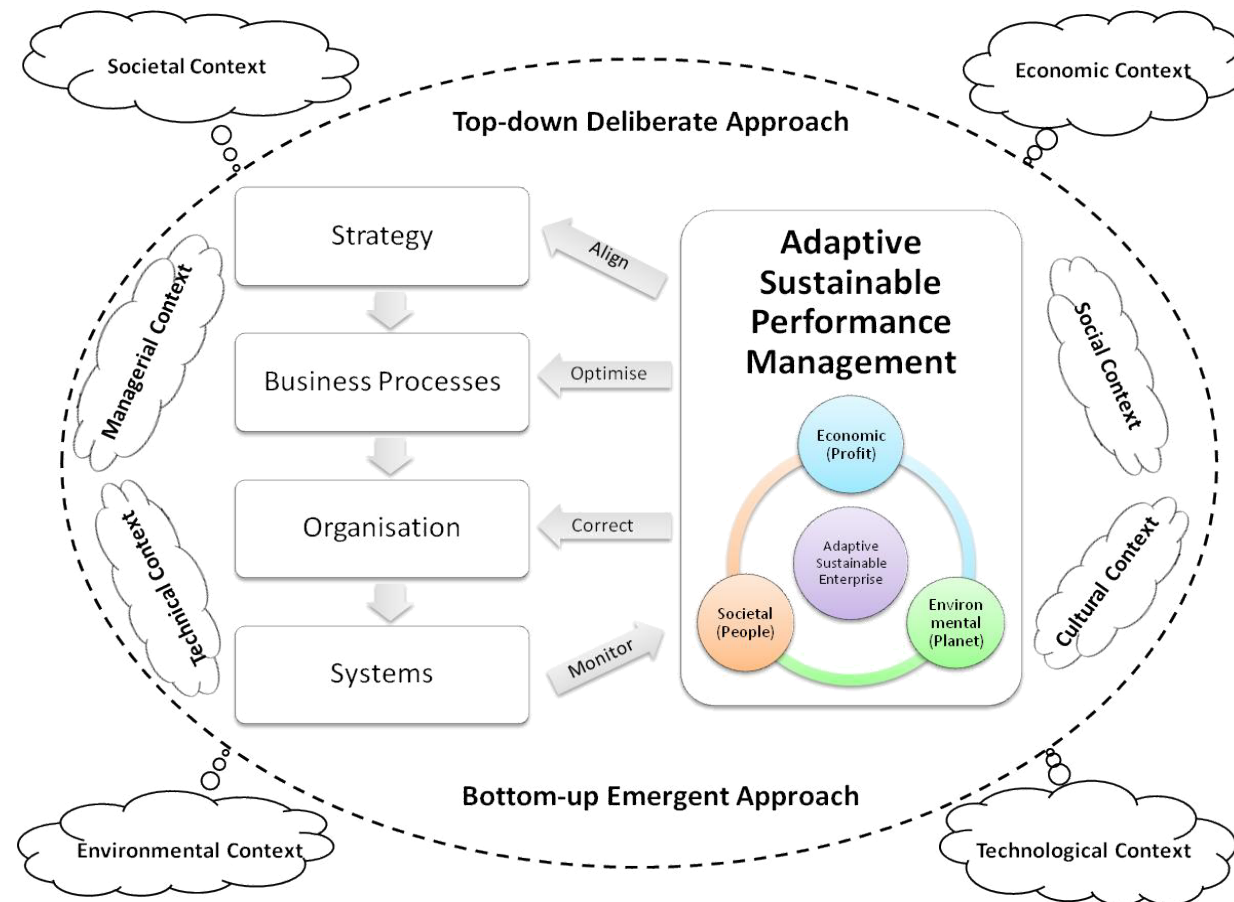
- H2a: The relationship between SNT use and social competence will vary depending on an individual's personality (Ellison et al 2007, Ross et al 2009)
- H2b: The relationship between SNT use and social competence will vary depending on gender (Teo and Lim 2000, Gross, 2004)



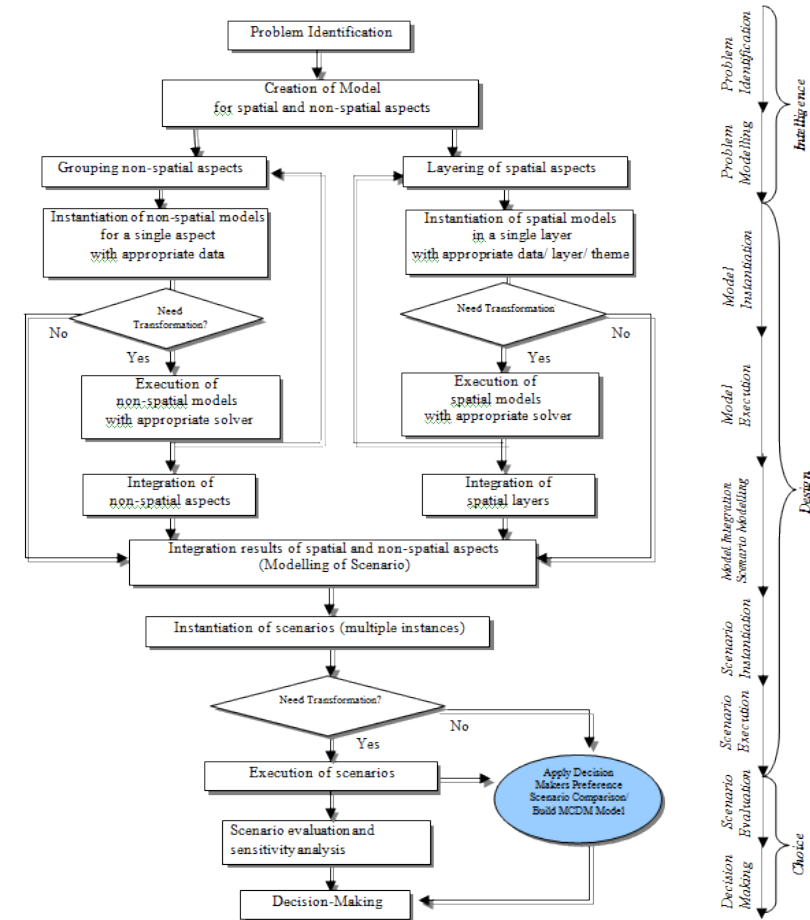
Models Hypotheses **Instruments**

- Hendrik's Delphi Instrument – Survey Monkey
- Hendrik's Survey Instrument – Survey Monkey
- Shahper's Survey Instrument – Survey Monkey
- Valeria's Survey Instrument – Survey Monkey
- Gabrielle's Delphi Instrument – Survey Monkey

Processes Methods Algorithms

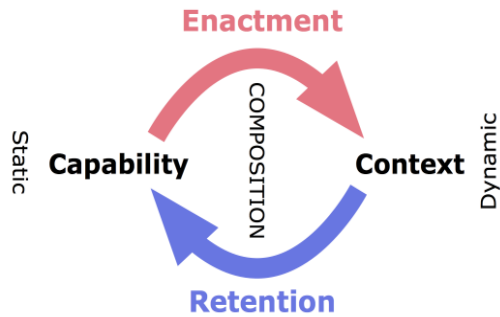


Gabrielle

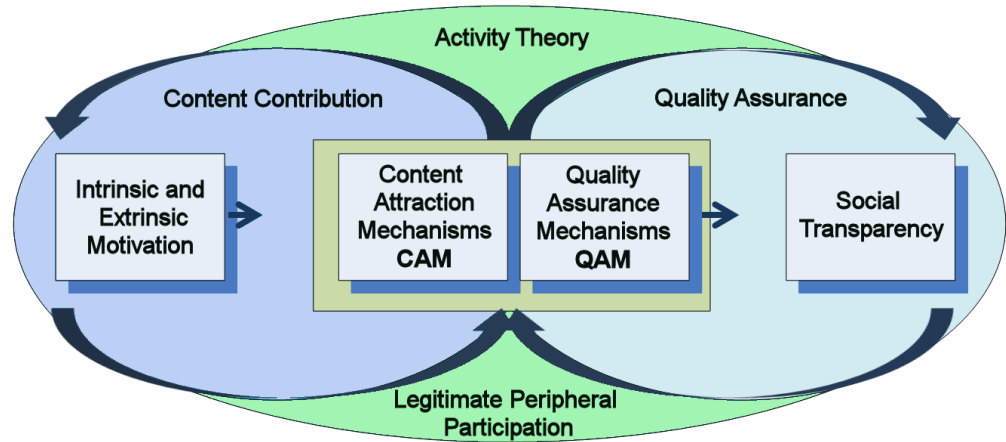


Shan

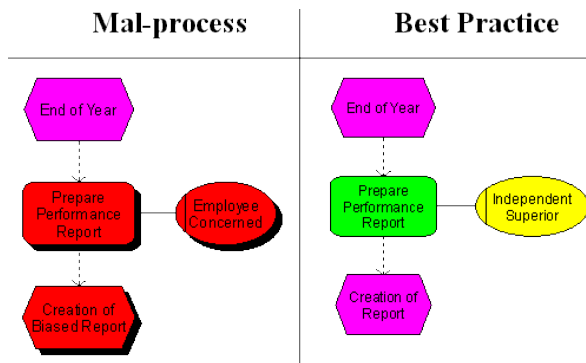
Processes Methods Algorithms



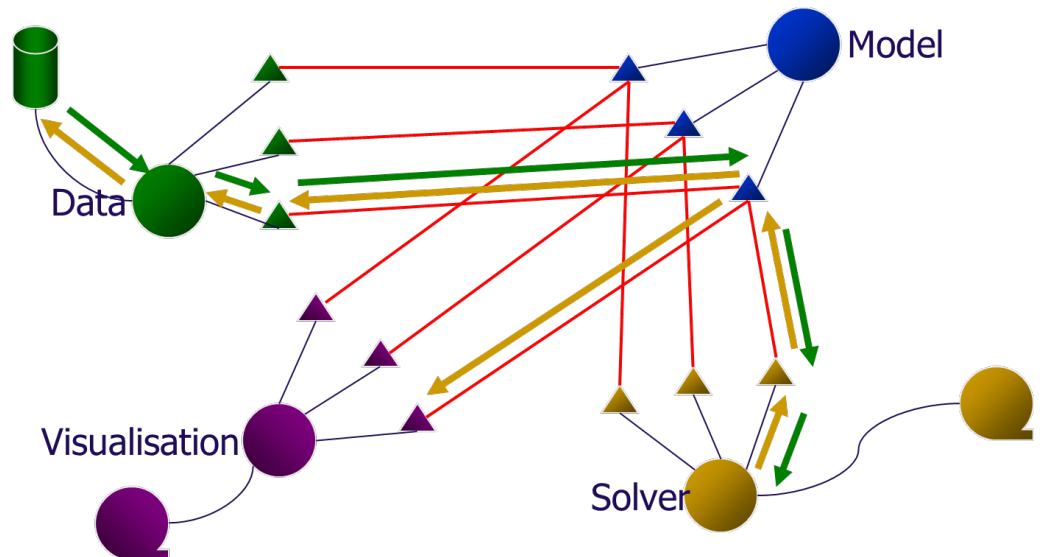
Max



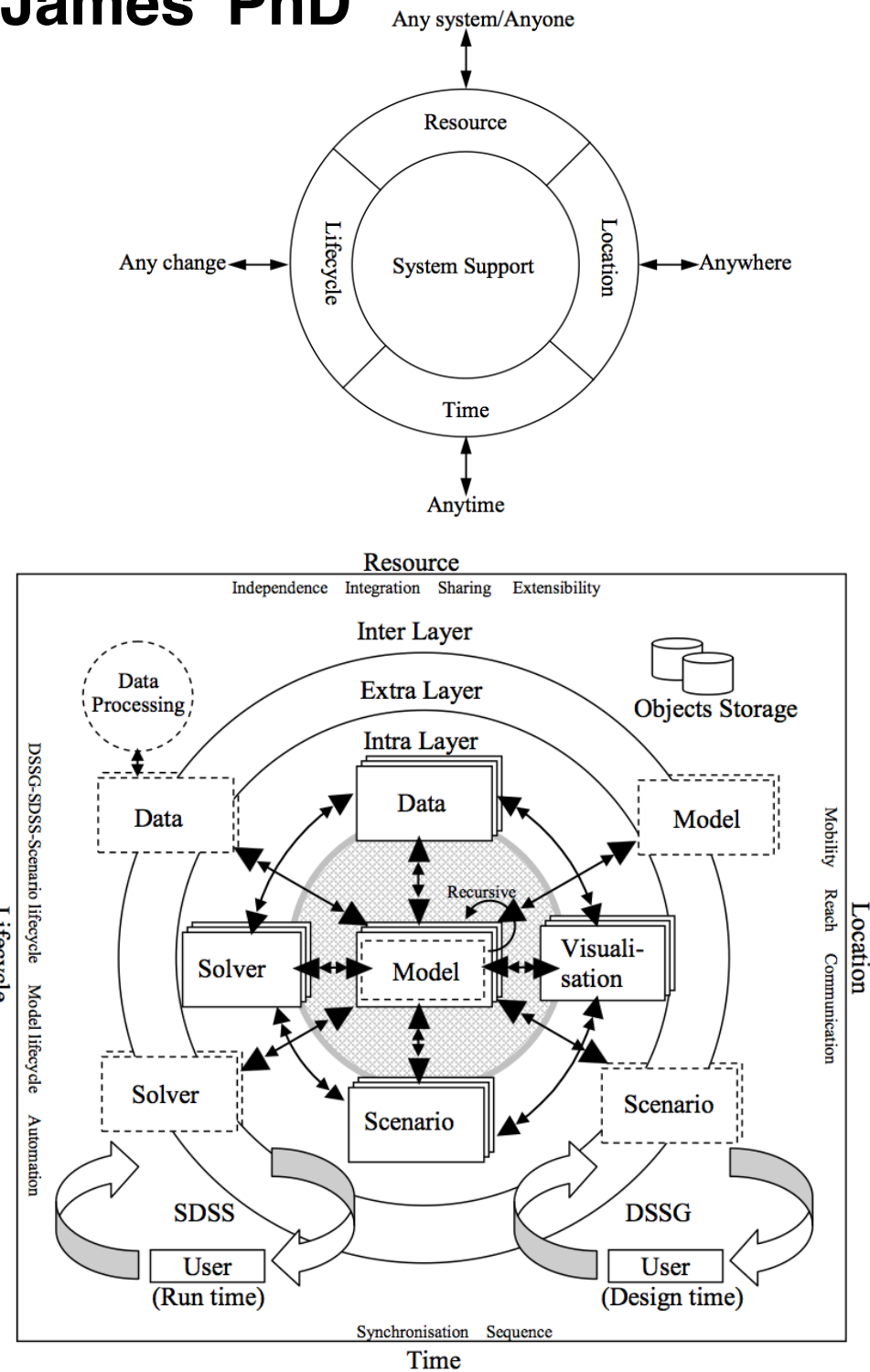
Tomi



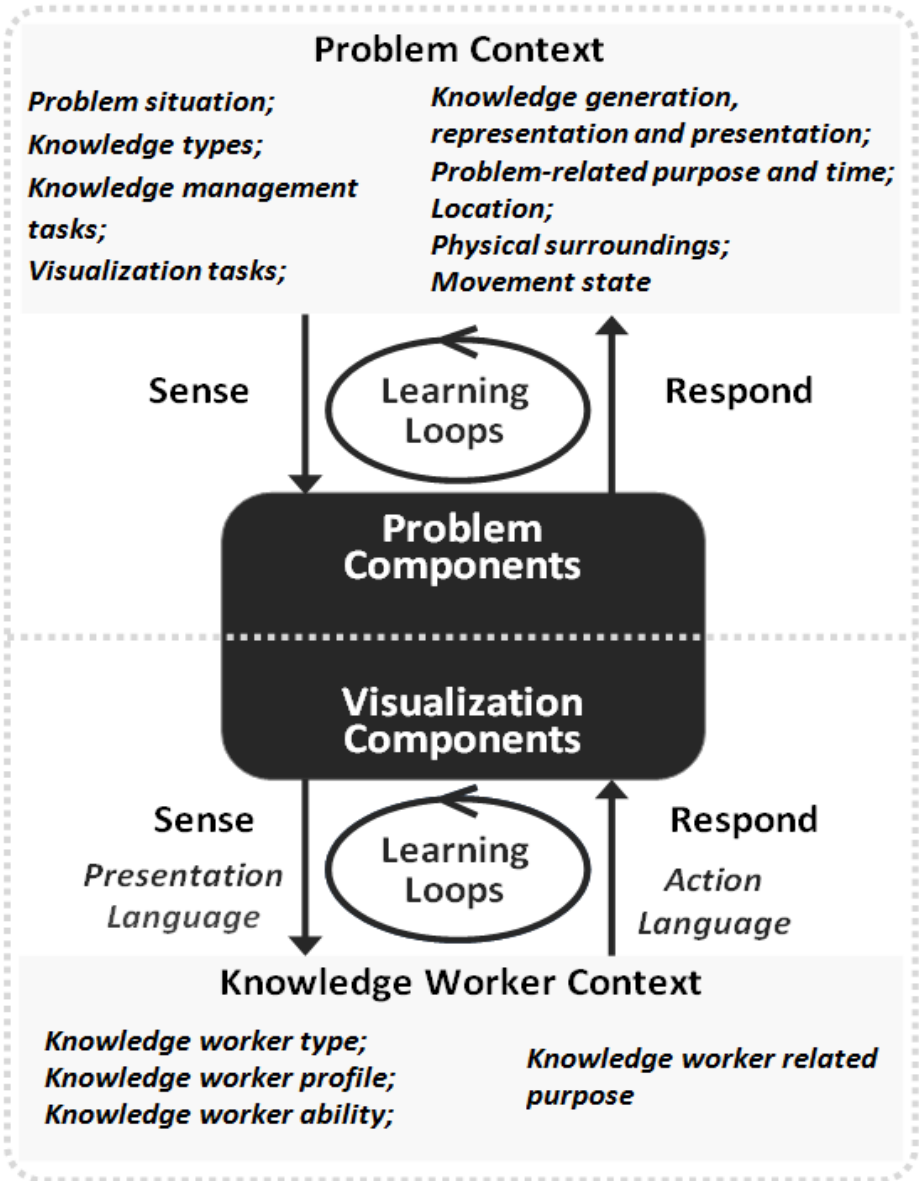
Victor



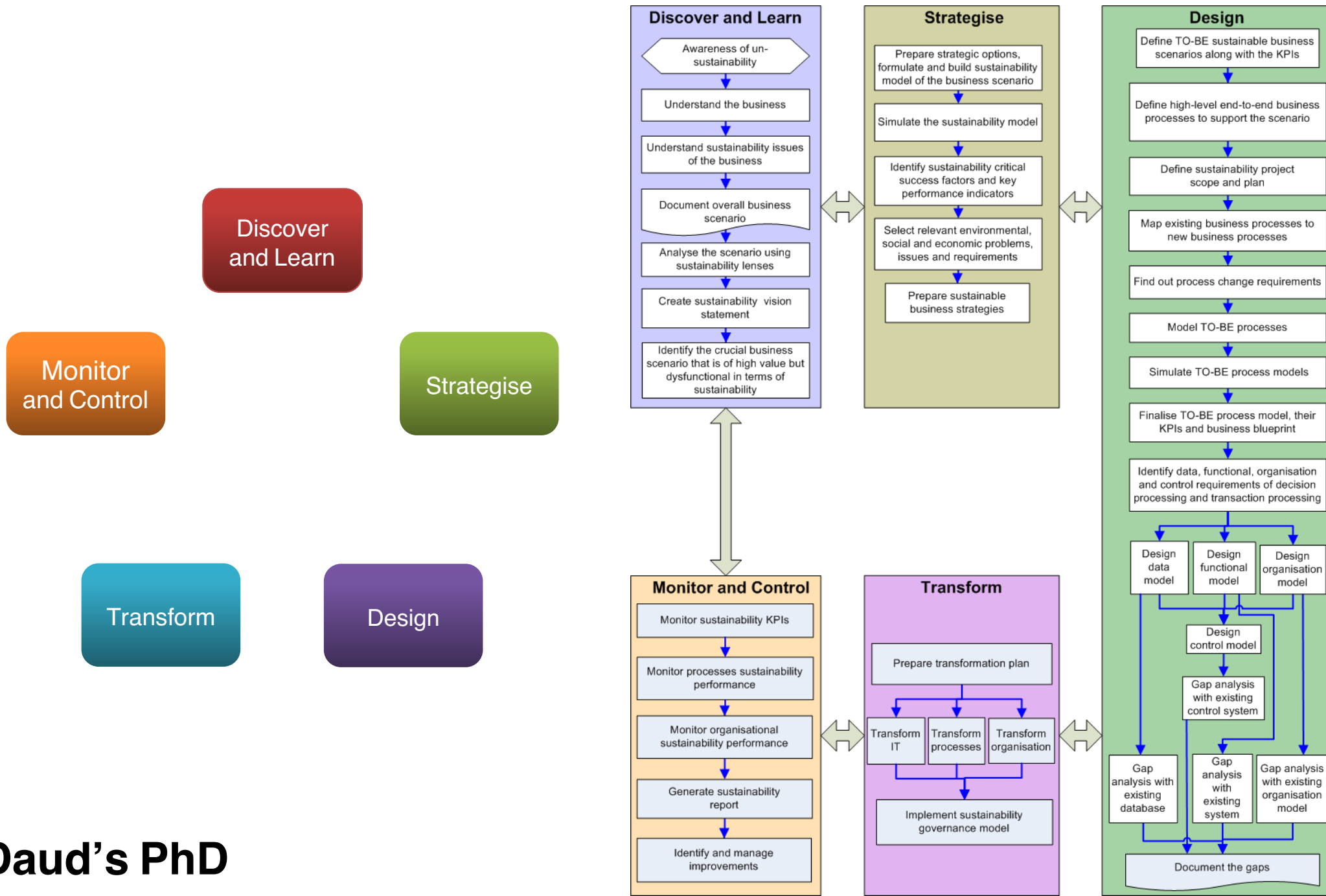
Sarah, Stephen, Doug



Frameworks

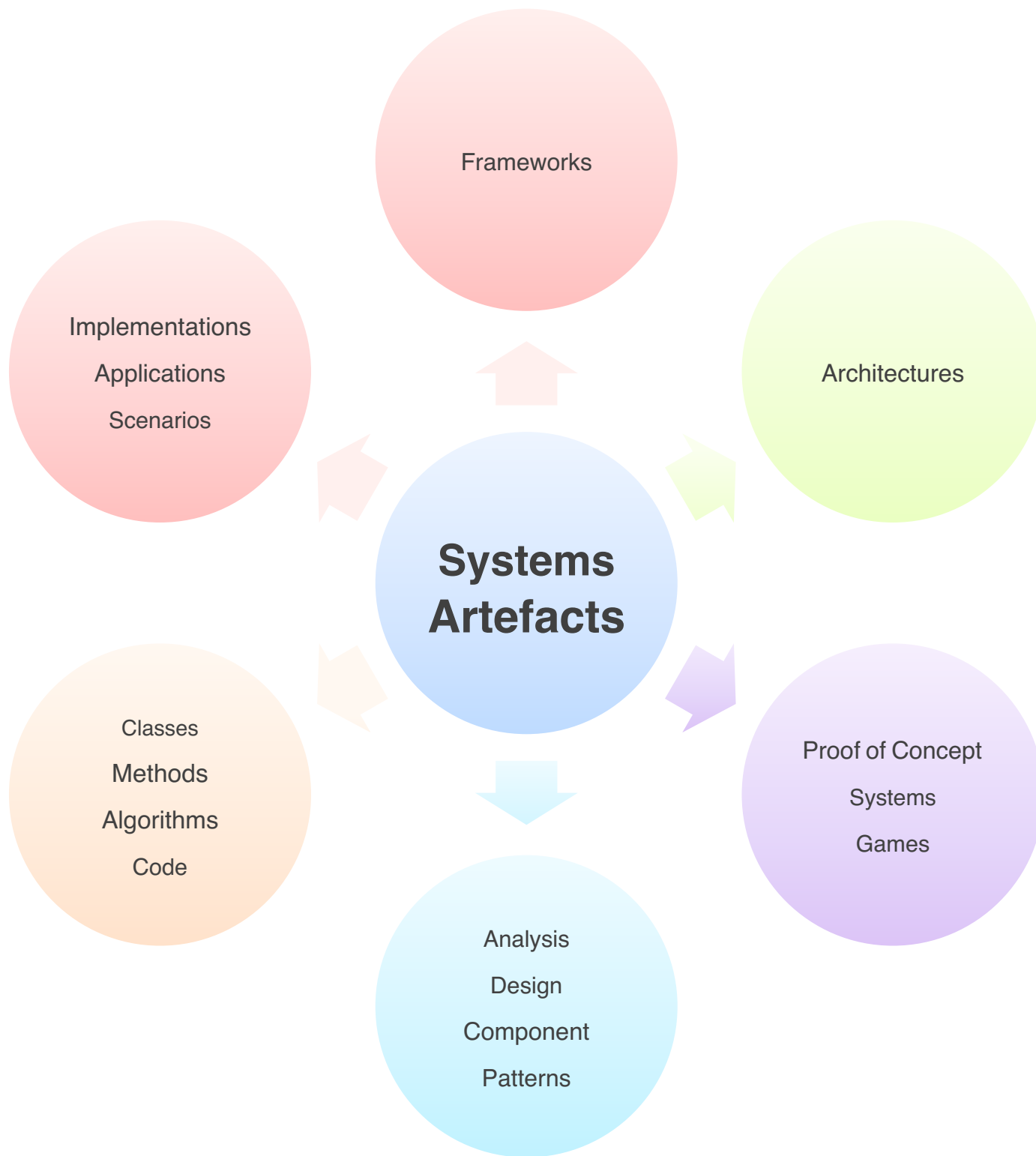


Roadmaps



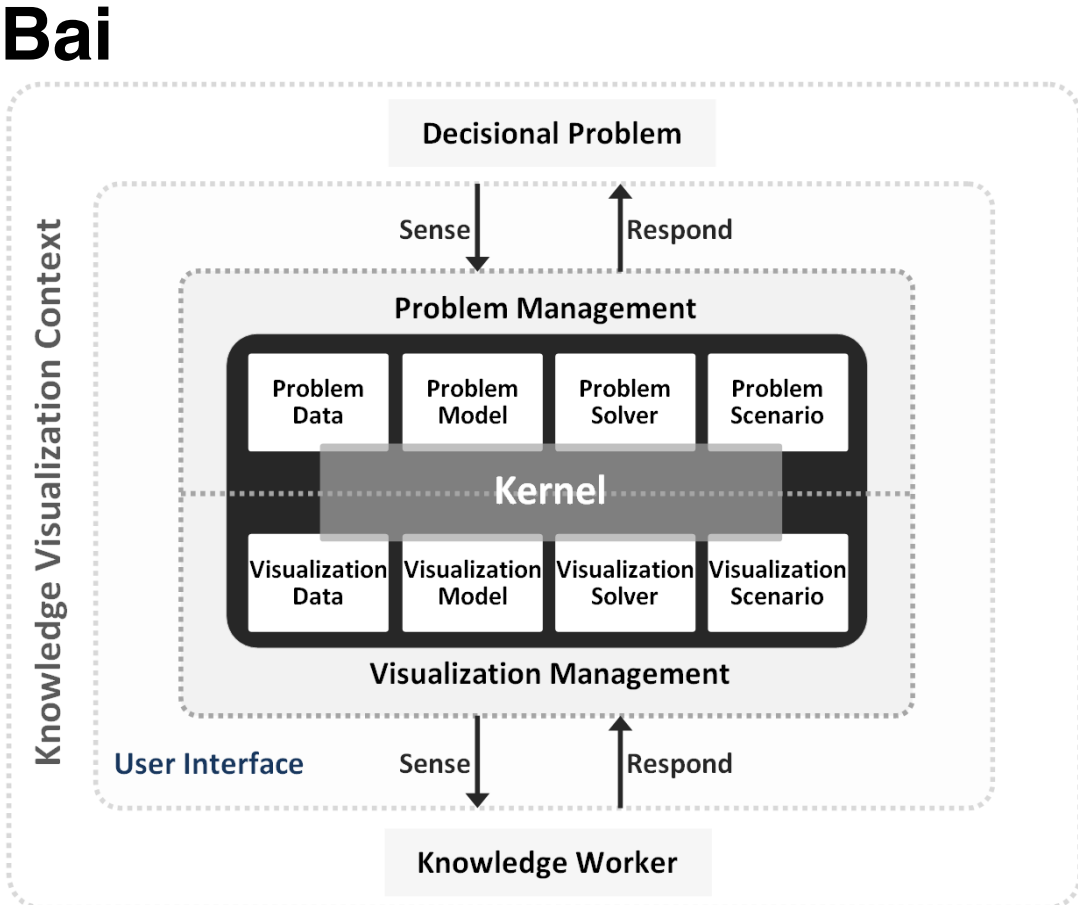
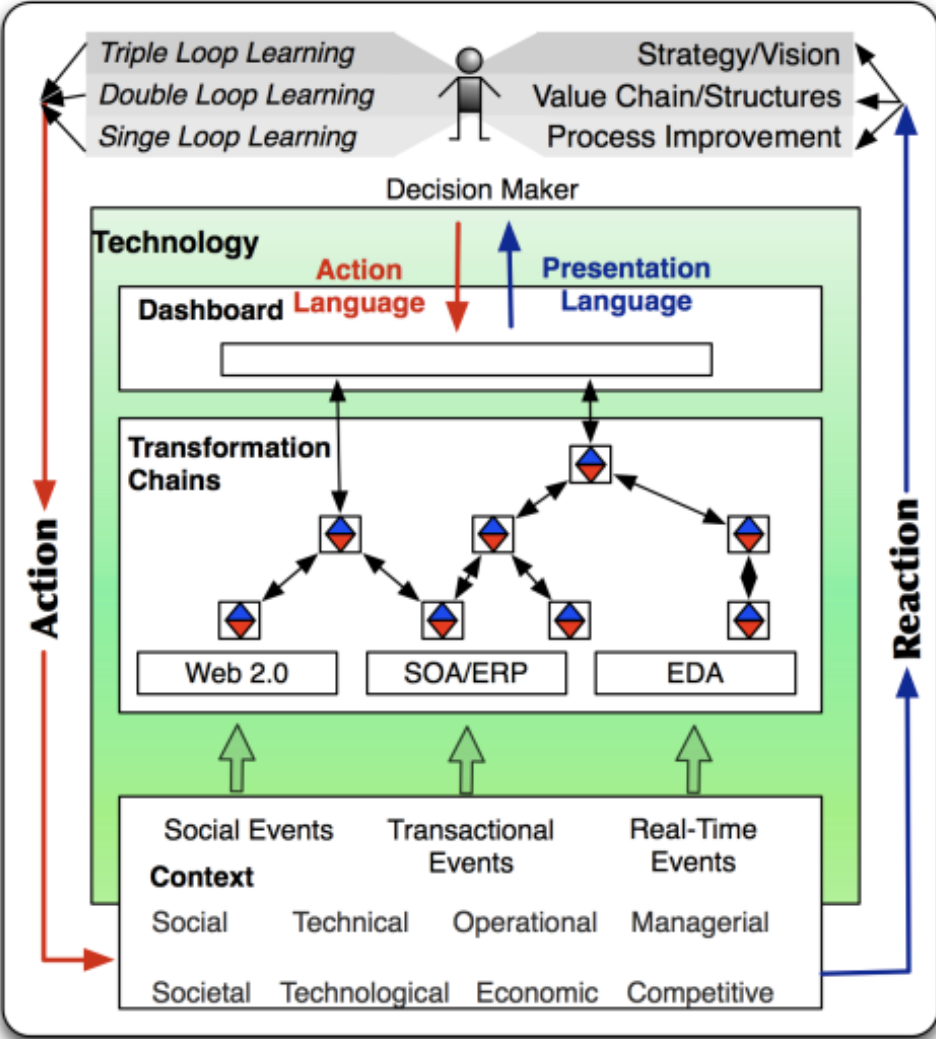
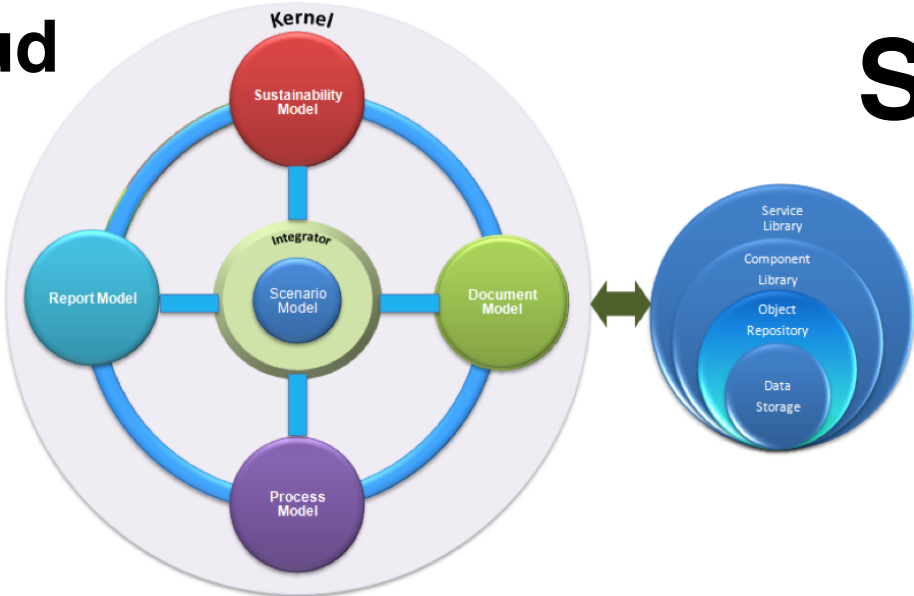
Design – System Artefacts

- Design
- Develop
- Proof of concept
- Prototypes
- Wireframes
- Breadboards
- Implement
- Apply in particular contexts – health, finance, organisations
- SMACT



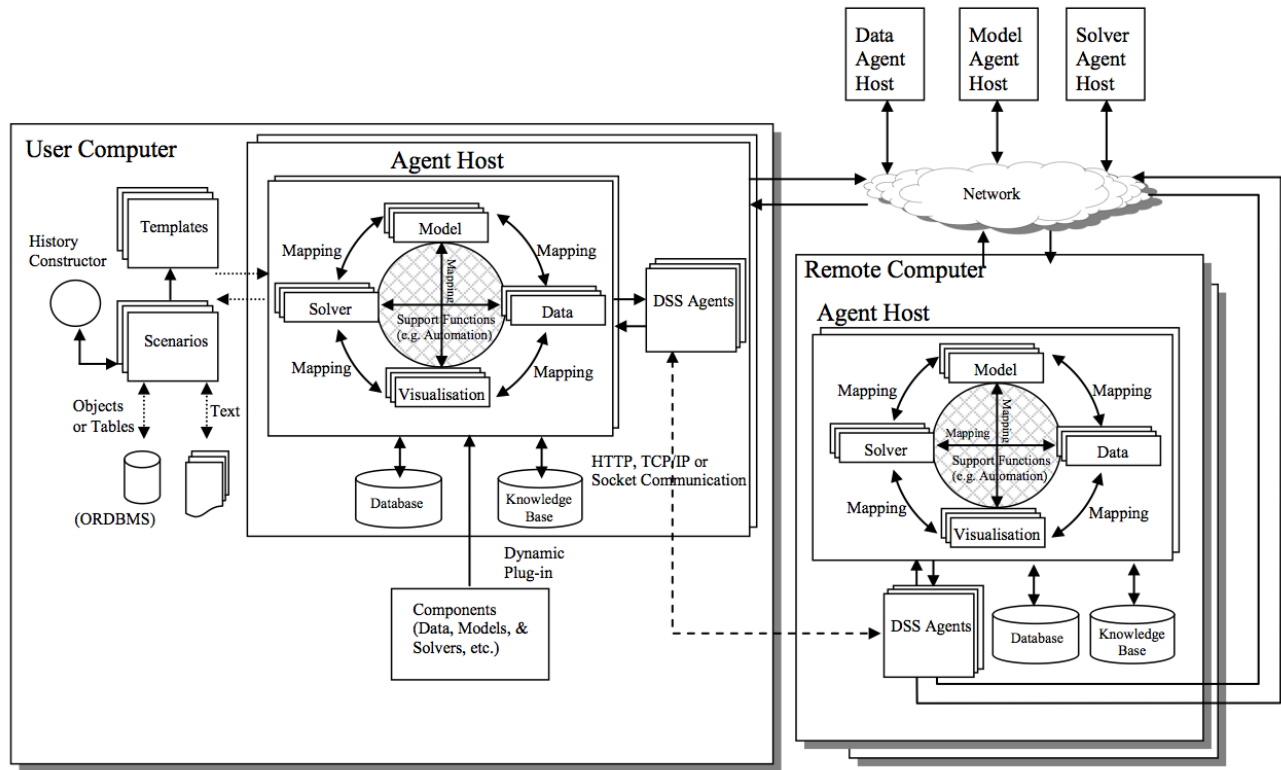
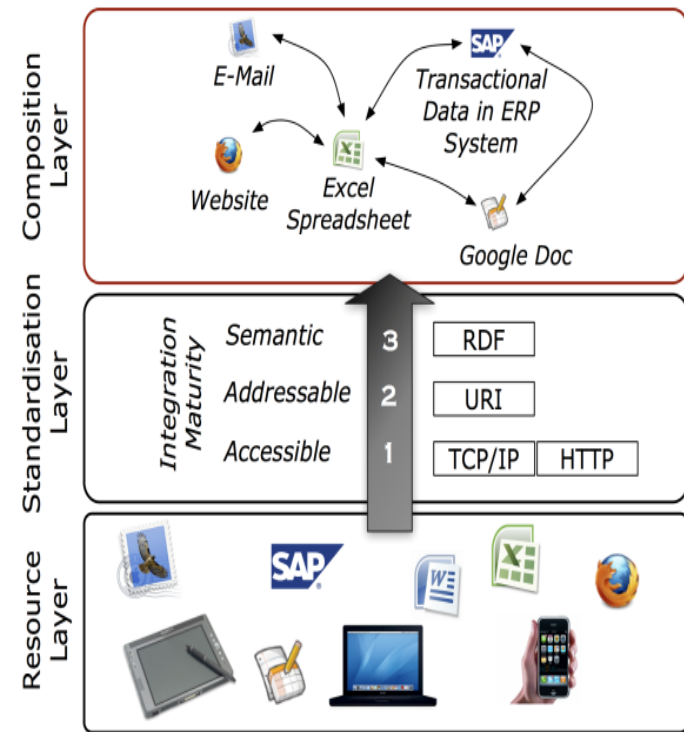
Daud

System Frameworks

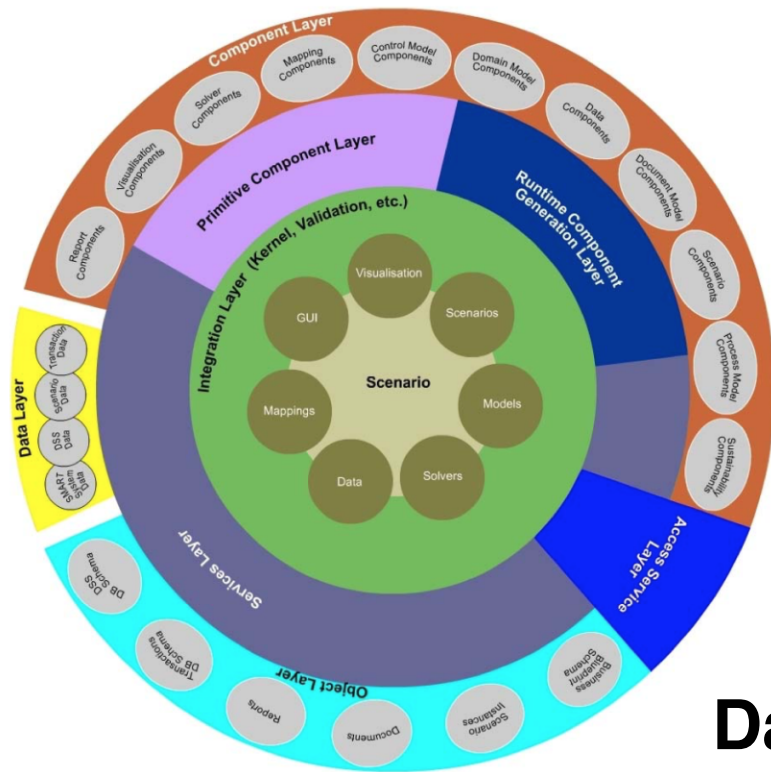


Max, Gabrielle, James

Max System Architectures



James' PhD



Daud's PhD

Proof of Concept Systems Games

Component Selection | **Spatial Manager** | Scenario Manager | Visualisation | Knowledge Manager | Component Manager

Model Distance

AddP1 AddP2 Reset

X1	Y1	X2	Y2
150	200	523	242

Data - Model **3128m**

Model-Solver Distance(toLayer)

Visualisation **AllDistance**

DSS Data

ctNo	Size	NoOfRoom	Style	YearOfBuild
0028	280	3	Flat	1978
0019	350	3	Flat	1980
0014	150	3	Flat	1970

Spatial Object

Spatial Themes

3
4
Runing Theme 1
Runing Theme 2

Show Themes
Theme Manager

Spatial Layers

Path
Racing Event
Restaurant
School
Shop
Toilet
TrafficLight

School
Shop

Show Layers
Layer Manager

Spatial Objects

Spatial Object Name **Hospital 2**

Spatial Object Type **Hospital**

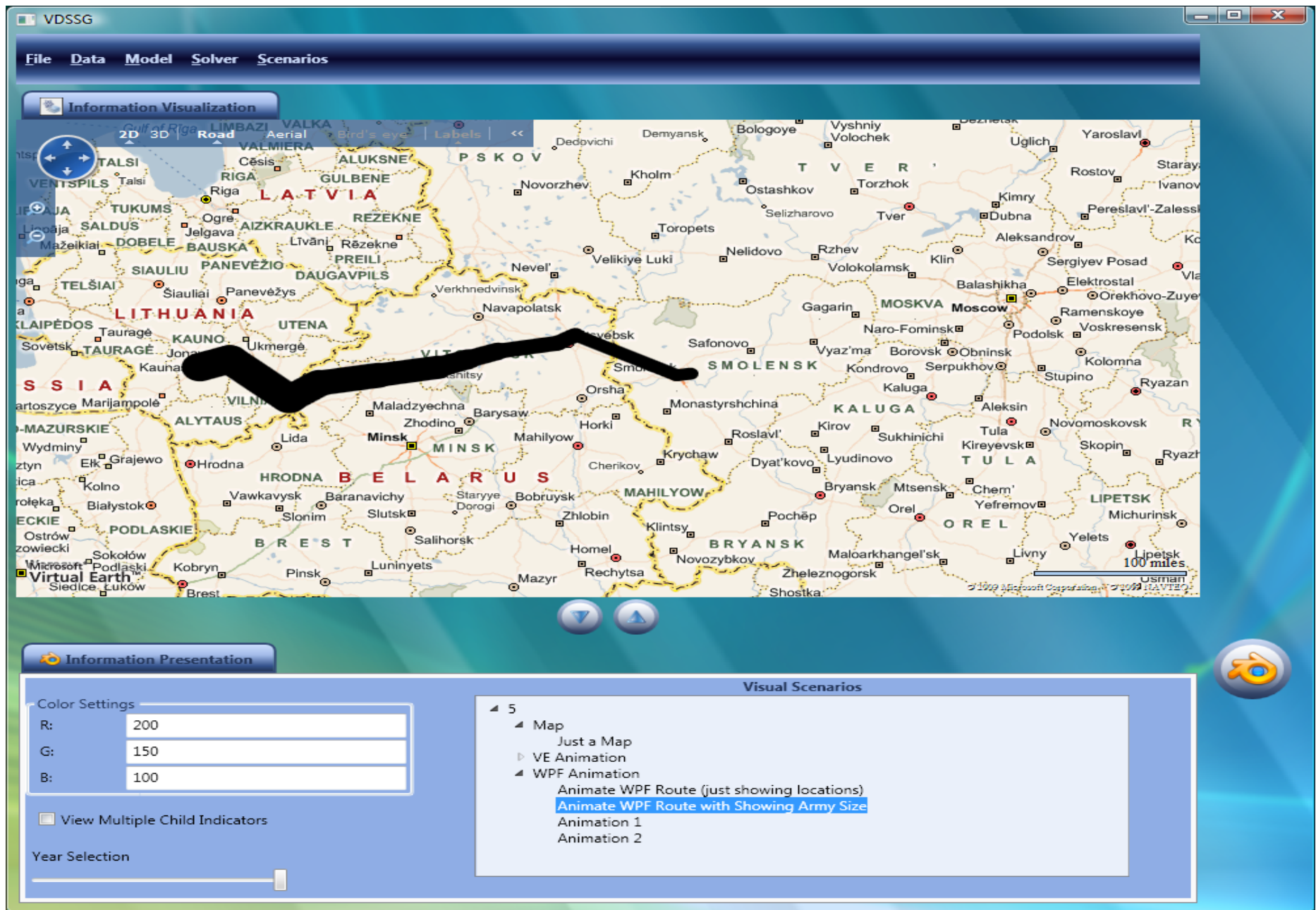
DrinkPoint4
Hospital 1
Hospital 2
Mt Albert
New Windsor
Oval2

523 242

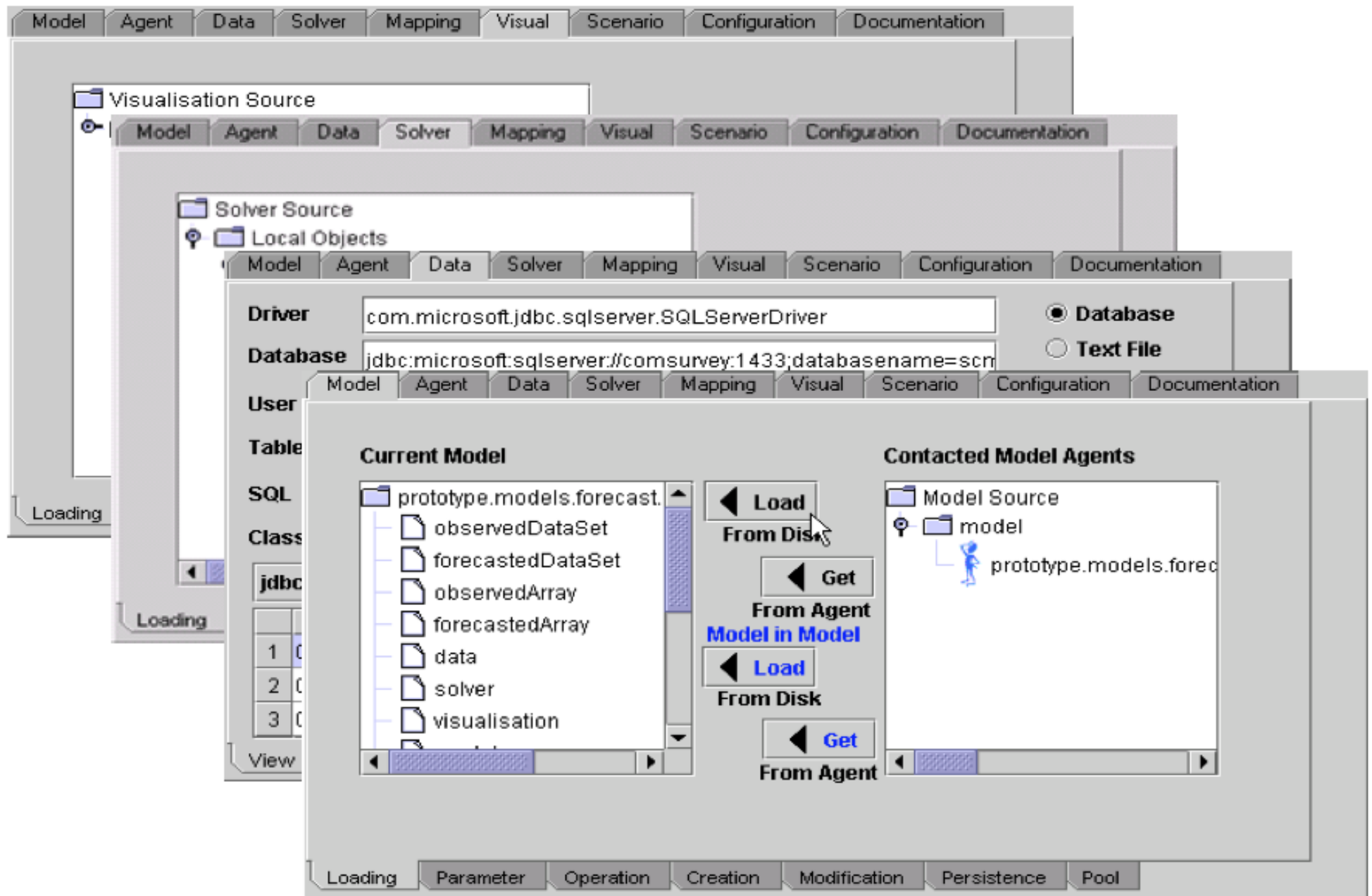
Numbers of Selected **3**

Instance 1 of 45

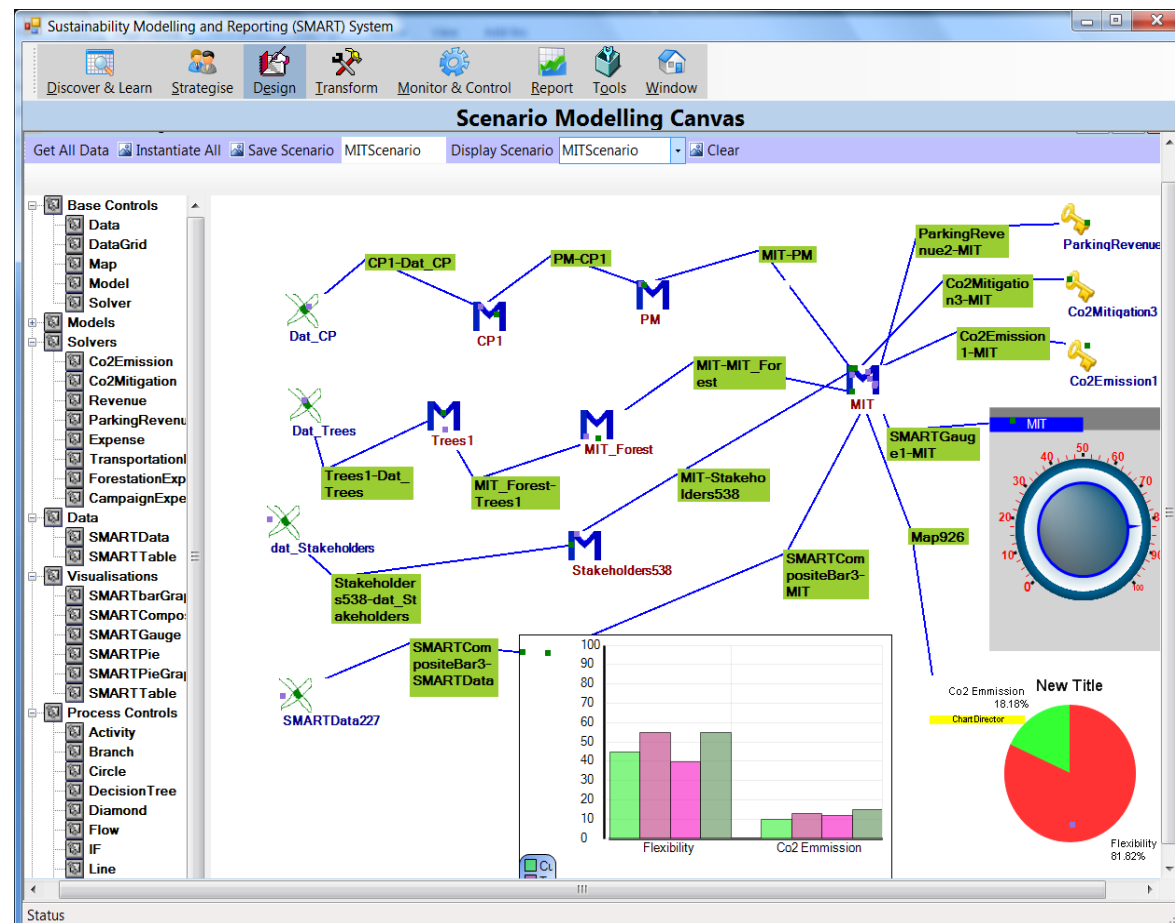
Proof of Concept Systems Games



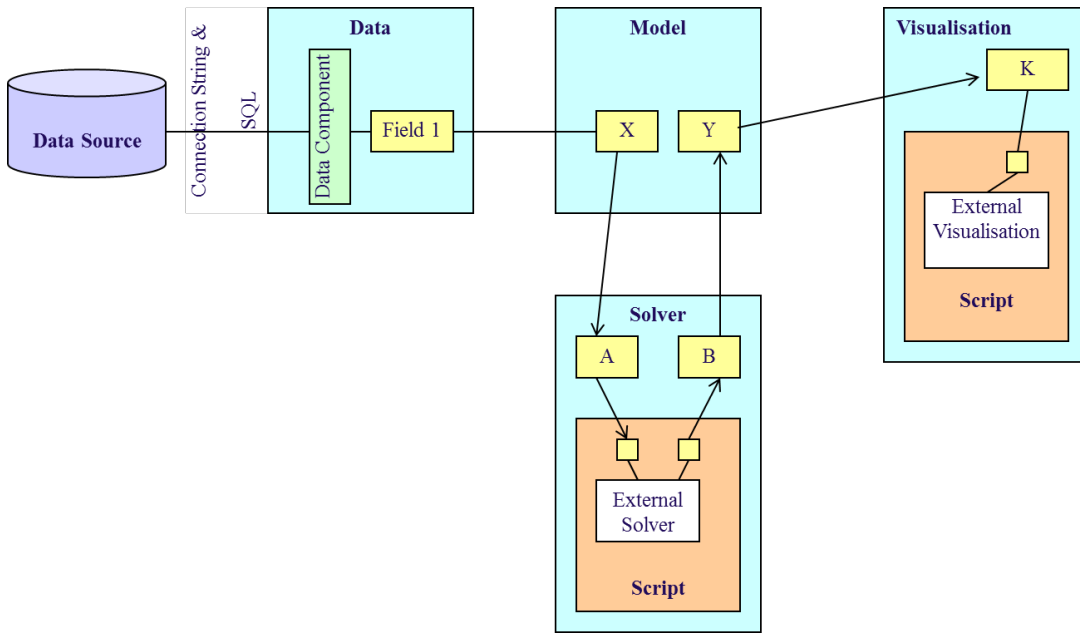
Proof of Concept Systems Games



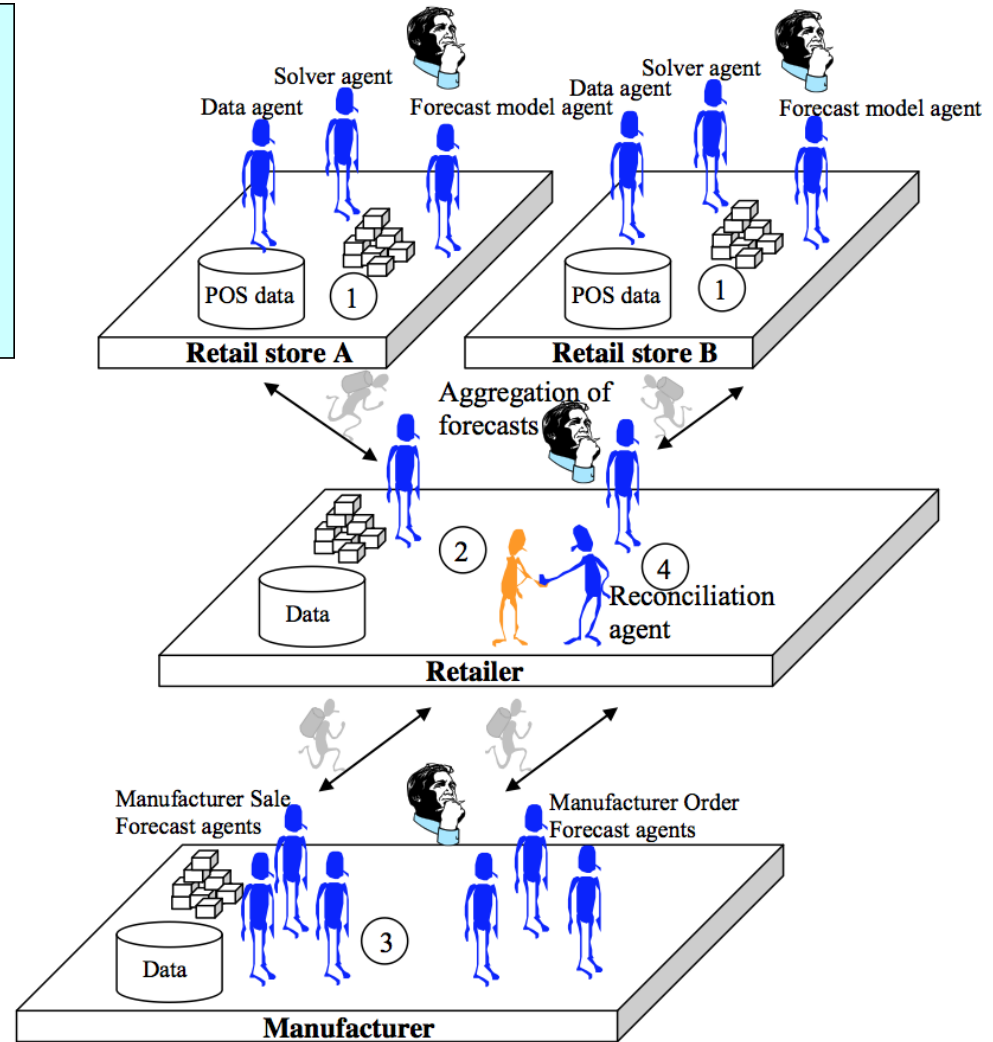
Proof of Concept Systems Games



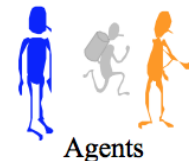
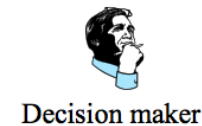
Analysis Design Component Patterns



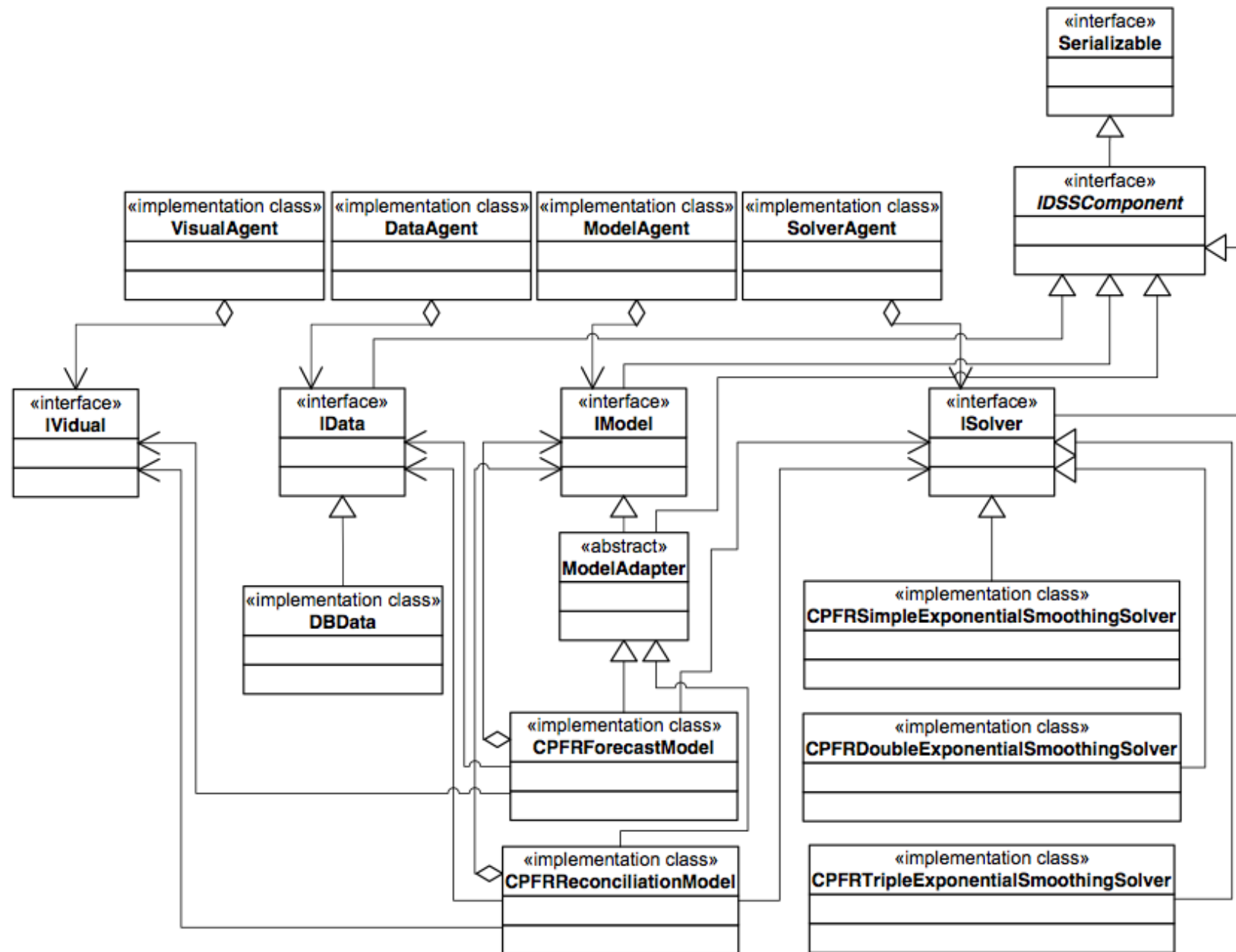
Sarah, Stephen, Doug



Legend



Classes Methods Algorithms Code

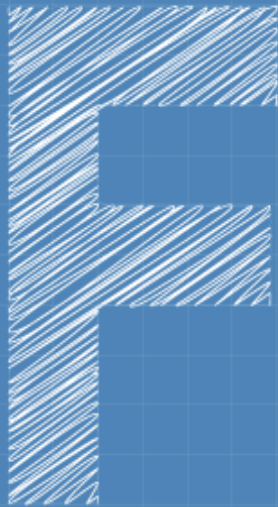


Implementations Applications Scenarios

- Shan's spatial decision making scenarios
- James' CPFR scenario
- Asfahaan's Te Reo Maori Implementation
- Rad's CHH applications
- Daud's parking, copier,.. scenarios
- Bai's Napoleon's March, Vector,.. scenarios
- Valeria's financial and health social networks
- Leila's tourism scenarios
- Claris' shopping and chronic disease scenarios

Evaluate

- FURPS
- Effective and efficient
- Hevner's Guidelines
- Experiments/Simulations
- Case studies/Field Studies
- Surveys/Delphi Studies/Interviews
- Experience
- FEDS



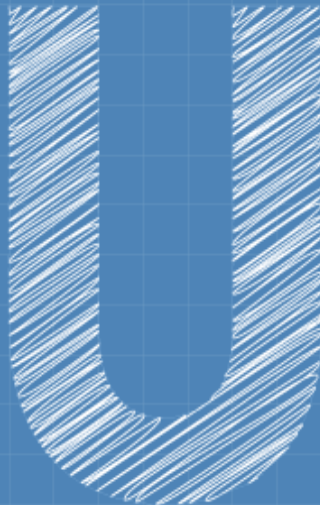
FUNCTIONALITY

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USABILITY

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RELIABILITY

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PERFORMANCE

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SUPPORTABILITY

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Hevner et al (2004) Evaluation

Table 2. Design Evaluation Methods

1. Observational	Case Study: Study artifact in depth in business environment
	Field Study: Monitor use of artifact in multiple projects
2. Analytical	Static Analysis: Examine structure of artifact for static qualities (e.g., complexity)
	Architecture Analysis: Study fit of artifact into technical IS architecture
	Optimization: Demonstrate inherent optimal properties of artifact or provide optimality bounds on artifact behavior
	Dynamic Analysis: Study artifact in use for dynamic qualities (e.g., performance)
3. Experimental	Controlled Experiment: Study artifact in controlled environment for qualities (e.g., usability)
	Simulation – Execute artifact with artificial data
4. Testing	Functional (Black Box) Testing: Execute artifact interfaces to discover failures and identify defects
	Structural (White Box) Testing: Perform coverage testing of some metric (e.g., execution paths) in the artifact implementation
5. Descriptive	Informed Argument: Use information from the knowledge base (e.g., relevant research) to build a convincing argument for the artifact's utility
	Scenarios: Construct detailed scenarios around the artifact to demonstrate its utility

Reflect **Learn** Generalise **Communicate**

- Write a **research paper**:
 - Title
 - Abstract (150 words) and 6 keywords
 - Introduction
 - Observe using SLR and SSR
 - Theorise Conceptual Artefacts
 - Design System Artefacts
 - Evaluate
 - Discuss – Reflect, Learn, Generalise
 - Conclude
 - 10-20 pages (single space, 11 font size, with normal margins, [APA](#) format)
 - Refer to journals such as [Decision Support Systems](#) and conferences such as [HICSS](#)

Case: Students Pandemic Stress, Fears, and Mental Health

Pandemic stress: The toll it's taking on students

College students' stress and health in the COVID-19 pandemic: The role of academic workload, separation from school, and fears of contagion

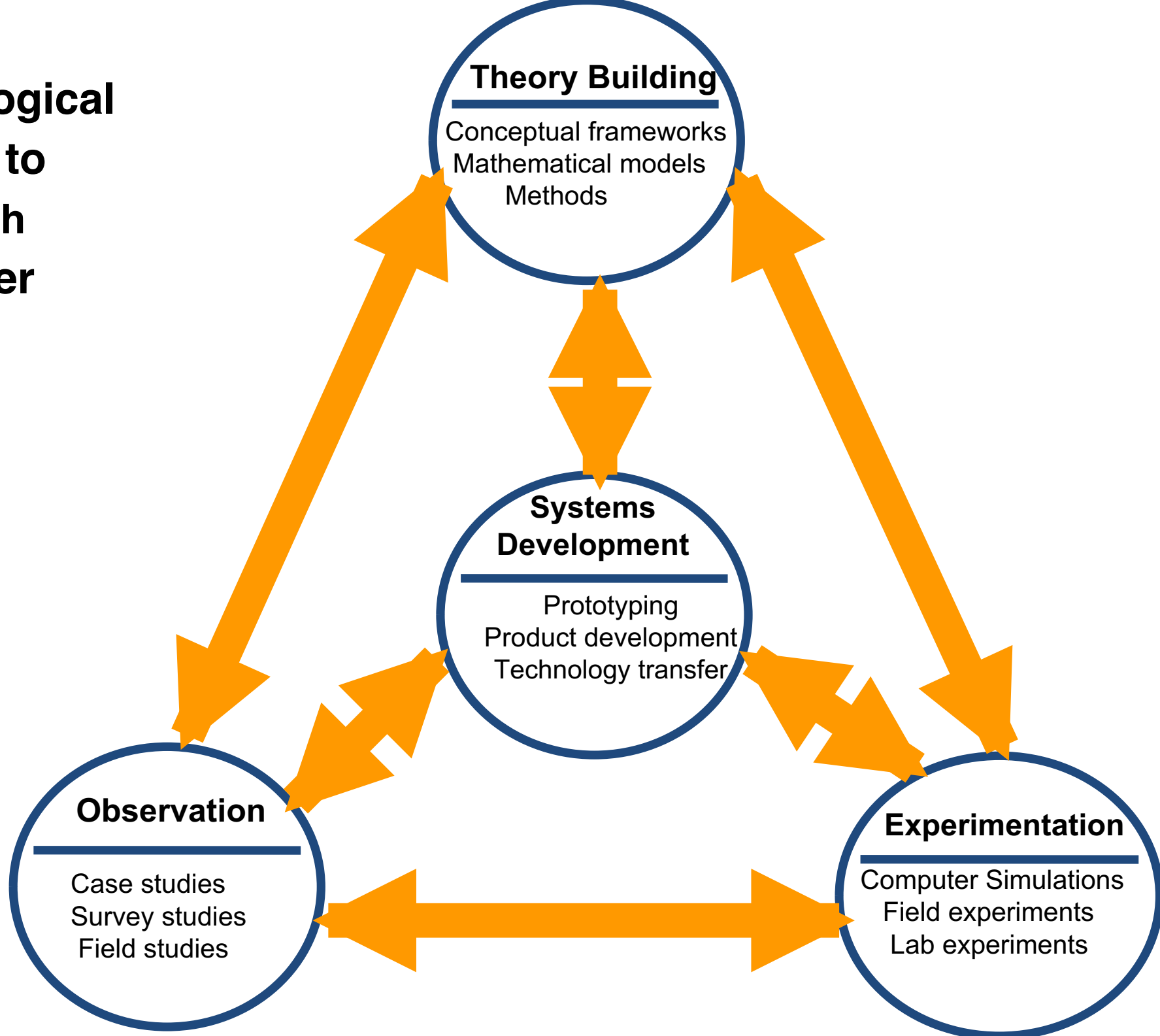
Mental Health Needs Rise With Pandemic

Models

Processes

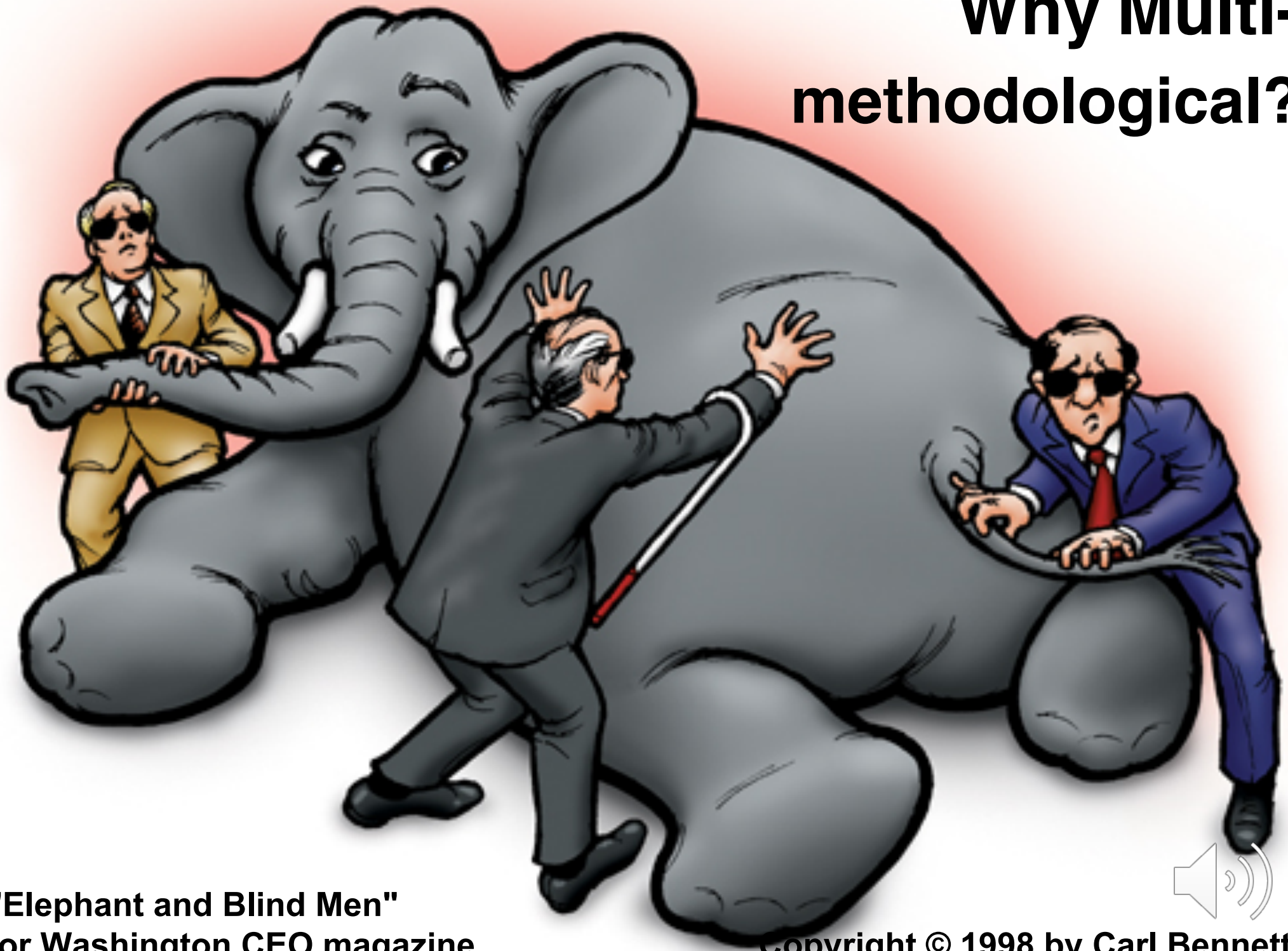
Ideas

A Multi- methodological Approach to IS Research Nunamaker et al. 1991



- **What** are you trying to do? Articulate your objectives using absolutely no jargon. (SMART & BHAG)
- **How** is it done today?
- **What** are the limits of current practice?
- **What's** new in your approach?
- **Why** do you think it will be successful?
- **Who** cares?
- If you're successful, **what** difference will it make?
- **What** are the risks and the payoffs?
- **How** much will it cost?
- **How** long will it take?
- **What** are the midterm and final "exams" to check for success?
- **What** is your contingency plan? *Rad Jrad*

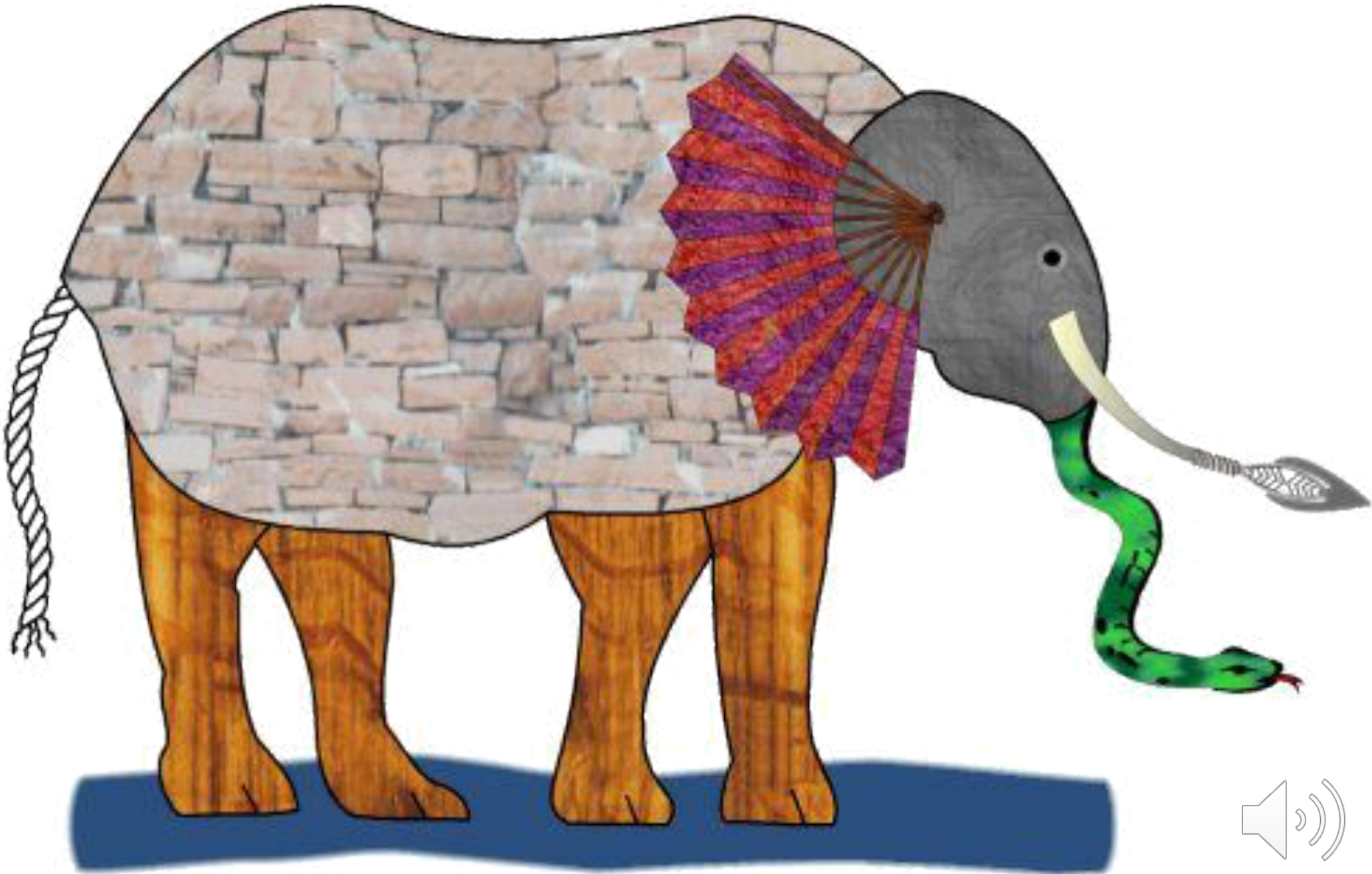
Why Multi-methodological?



"Elephant and Blind Men"
for Washington CEO magazine

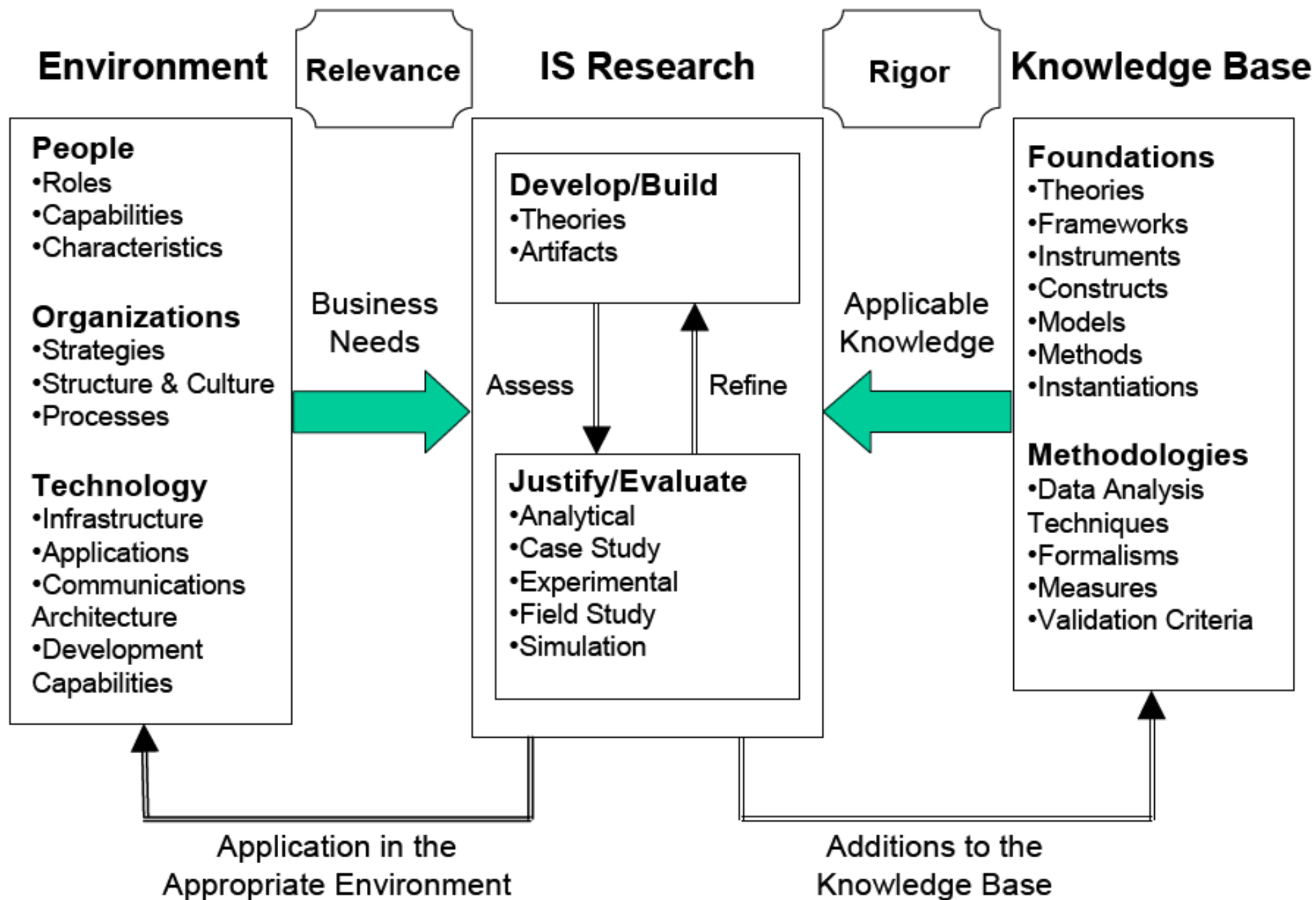
Copyright © 1998 by Carl Bennett.

Why Multi-methodological?



Picture by Paul Galdone, New York:Whittlesey House.

Hevner et al (2004) DSR Framework

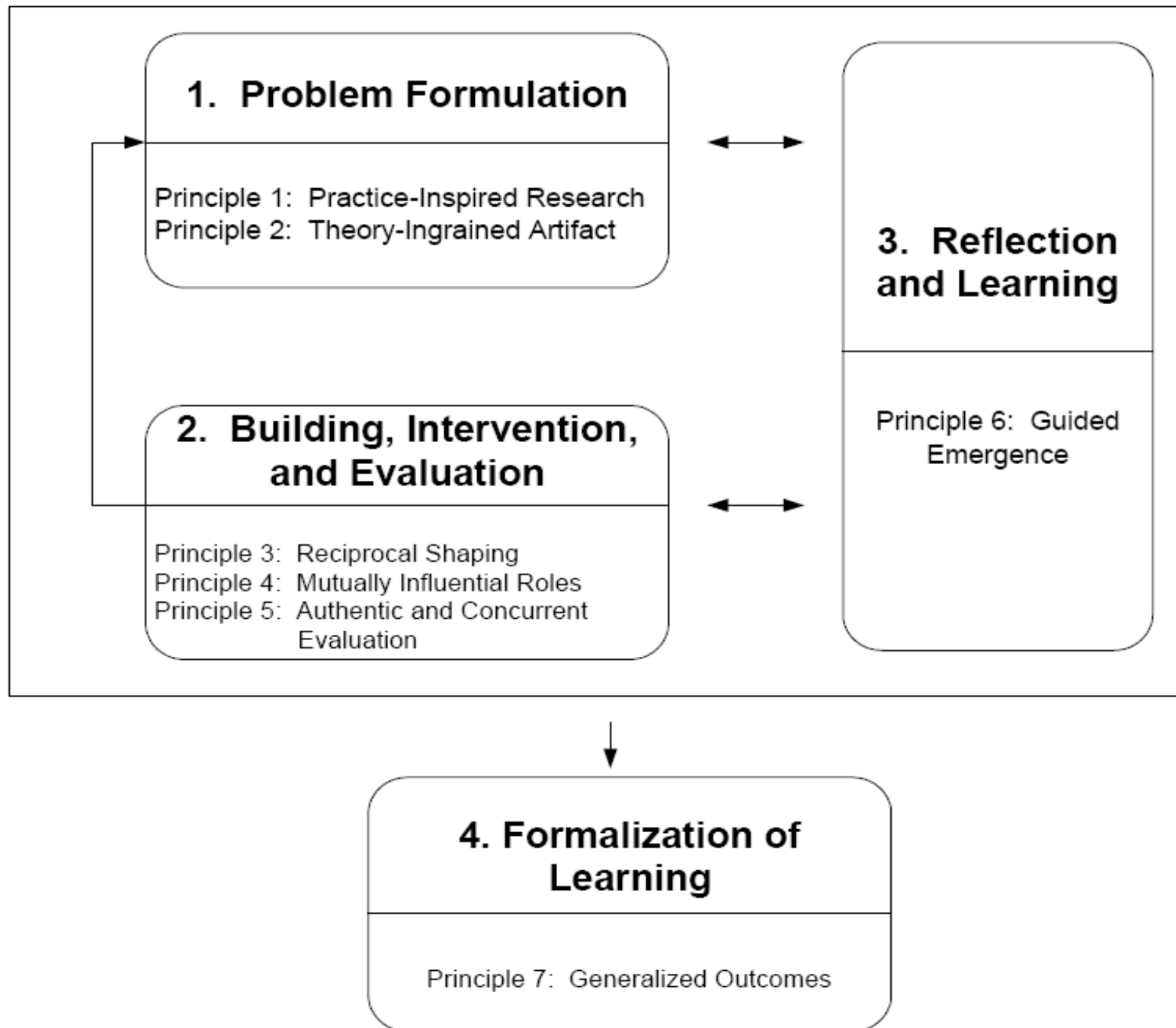


Hevner et al (2004) Guidelines

Table 1. Design-Science Research Guidelines

Guideline	Description
Guideline 1: Design as an Artifact	Design-science research must produce a viable artifact in the form of a construct, a model, a method, or an instantiation.
Guideline 2: Problem Relevance	The objective of design-science research is to develop technology-based solutions to important and relevant business problems.
Guideline 3: Design Evaluation	The utility, quality, and efficacy of a design artifact must be rigorously demonstrated via well-executed evaluation methods.
Guideline 4: Research Contributions	Effective design-science research must provide clear and verifiable contributions in the areas of the design artifact, design foundations, and/or design methodologies.
Guideline 5: Research Rigor	Design-science research relies upon the application of rigorous methods in both the construction and evaluation of the design artifact.
Guideline 6: Design as a Search Process	The search for an effective artifact requires utilizing available means to reach desired ends while satisfying laws in the problem environment.
Guideline 7: Communication of Research	Design-science research must be presented effectively both to technology-oriented as well as management-oriented audiences.

Action Design Research



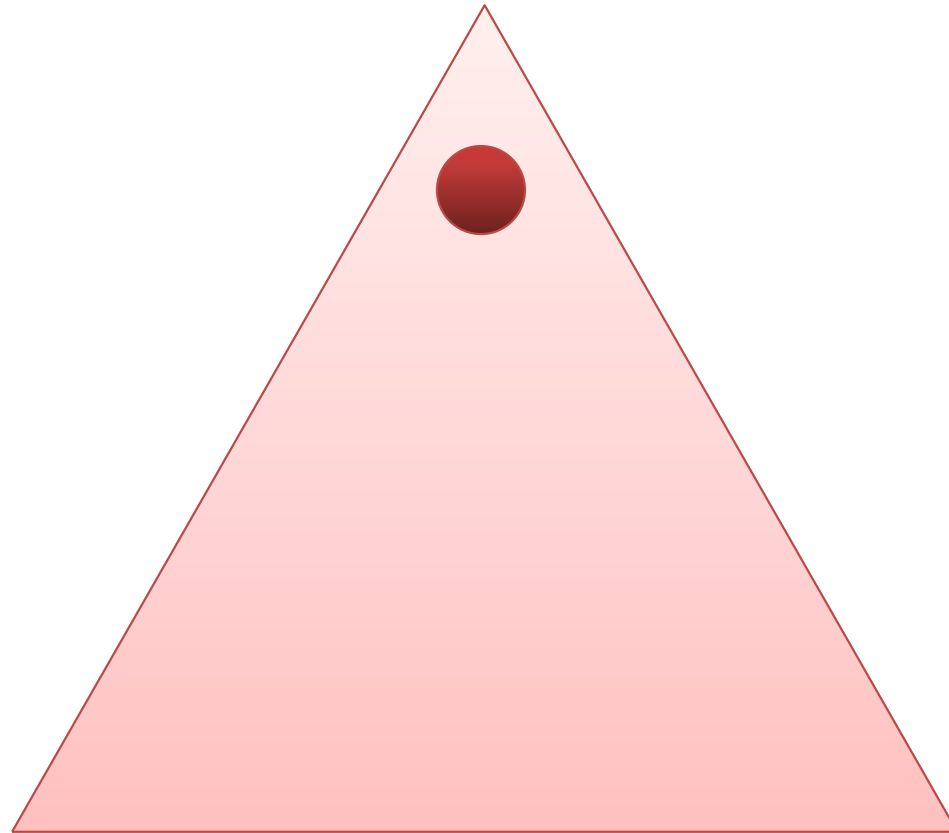
Remember the Triangle

“As a development manager you are working with only three things: Resources (people and money), features (product & its quality), and the Schedule (time). Changing one side of the triangle has an impact on at least one other side, usually two.”

McCarthy, 1995



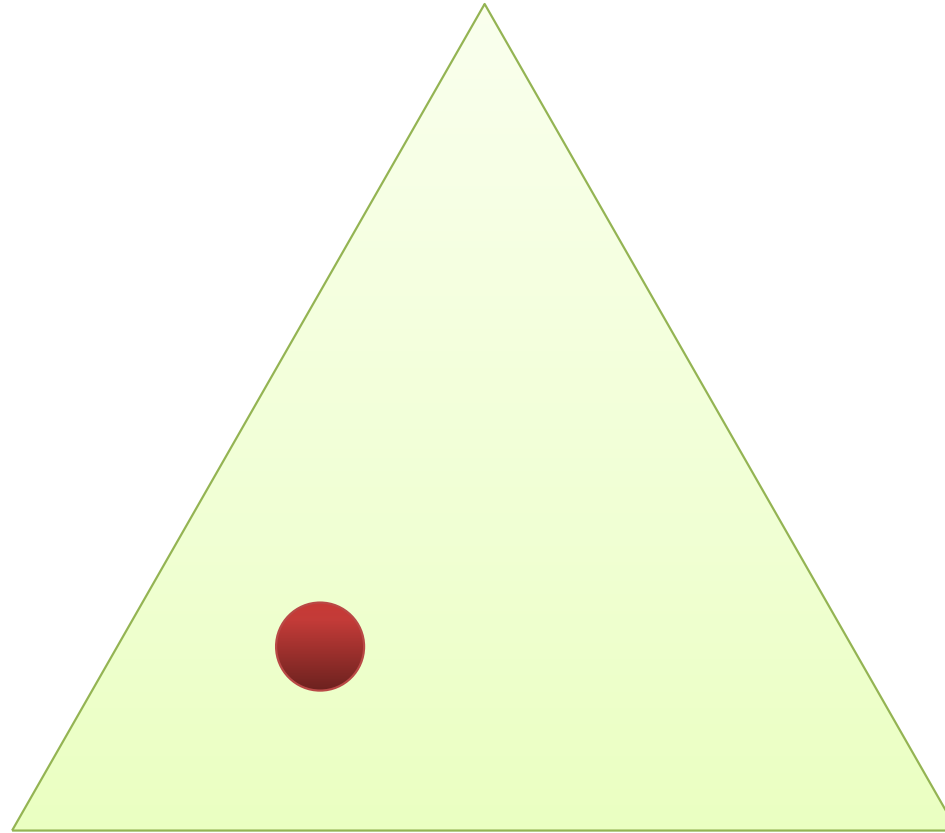
Quality



Effort

Time

Perfectionism



Quality

Pragmatism

Quality is not an act, it is a habit. *Aristotle*

Quality means doing it right when no one is looking. *Henry Ford*

The final output /
artefacts should be
good enough to show
to the CEO of the
company where you
want to be hired!

**A woman without her
man is nothing.**

**A woman, without her
man, is nothing.**

**A woman: without her,
man is nothing.**

A woman with dark hair is performing a magic trick. She is holding a lit candle between her hands, which are raised in front of her. The candle is glowing brightly, casting a warm light on her face and hands. She is wearing a light-colored, possibly white, top. In the foreground, there is a glass dish containing several dark, round objects, possibly pomegranates or apples, and a small glass jar. To the right, there is a dark, ornate object, possibly a lamp or a decorative piece. The background is dark, making the candle and the woman's face the central focus.

MINIMIZE

the

MAGIC

T[•] R A C

E A B I

L I T Y

G E N E R

A L I S A B

I L I T Y

VALIDITY

In the South Seas there is a cargo cult of people. During the war they saw airplanes land with lots of good materials, and they want the same thing to happen now. So they've arranged to make things like runways, to put fires along the sides of the runways, to make a wooden hut for a man to sit in, with two wooden pieces on his head like headphones and bars of bamboo sticking out like antennas-- he's the controller--and they wait for the airplanes to land. They're doing everything right. The form is perfect. It looks exactly the way it looked before. But it doesn't work. No airplanes land. So I call these things cargo cult science, because they follow all the apparent precepts and forms of scientific investigation, but they're missing something essential, because the planes don't land.

Now it behooves me, of course, to tell you what they're missing. But it would be just about as difficult to explain to the South Sea Islanders how they have to arrange things so that they get some wealth in their system. It is not something simple like telling them how to improve the shapes of the earphones. But there is *one* feature I notice that is generally missing in cargo cult science. That is the idea that we all hope you have learned in studying science in school--we never explicitly say what this is, but just hope that you catch on by all the examples of scientific investigation. It is interesting, therefore, to bring it out now and speak of it explicitly. It's a kind of scientific integrity, a principle of scientific thought that corresponds to a kind of utter honesty--a kind of leaning over backwards.

For example, if you're doing an experiment, you should report everything that you think might make it invalid--not only what you think is right about it: other causes that could possibly explain your results; and things you thought of that you've eliminated by some other experiment, and how they worked--to make sure the other fellow can tell they have been eliminated, Details that could throw doubt on your interpretation must be given, if you know them. You must do the best you can--if you know anything at all wrong, or possibly wrong-- to explain it. If you make a theory, for example, and advertise it, or put it out, then you must also put down all the facts that disagree with it, as well as those that agree with it.

Scientific

Integrity

Utter

Honesty

NOT

CARGO CULT

SCIENCE

I USED TO HATE WRITING
ASSIGNMENTS, BUT NOW
I ENJOY THEM.



I REALIZED THAT THE
PURPOSE OF WRITING IS
TO INFLATE WEAK IDEAS,
OBSCURE POOR REASONING,
AND INHIBIT CLARITY.



WITH A LITTLE PRACTICE,
WRITING CAN BE AN
INTIMIDATING AND
IMPENETRABLE FOG!
WANT TO SEE MY BOOK
REPORT?



"THE DYNAMICS OF INTERBEING
AND MONOLOGICAL IMPERATIVES
IN DICK AND JANE: A STUDY
IN PSYCHIC TRANSRELATIONAL
GENDER MODES."



ACADEMIA,
HERE I
COME!

NOT!!! Tell a
SIMPLE and
good STORY

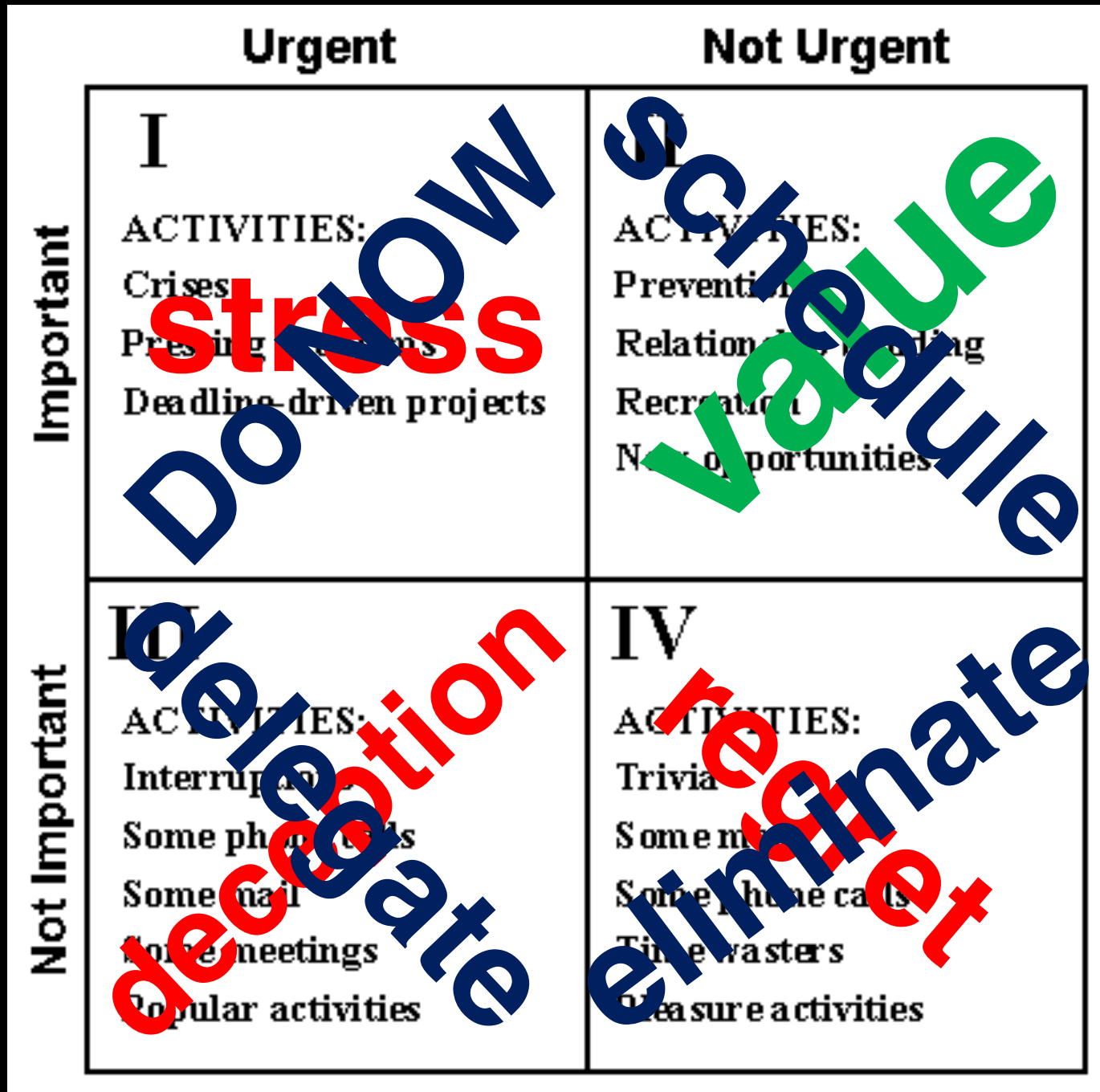


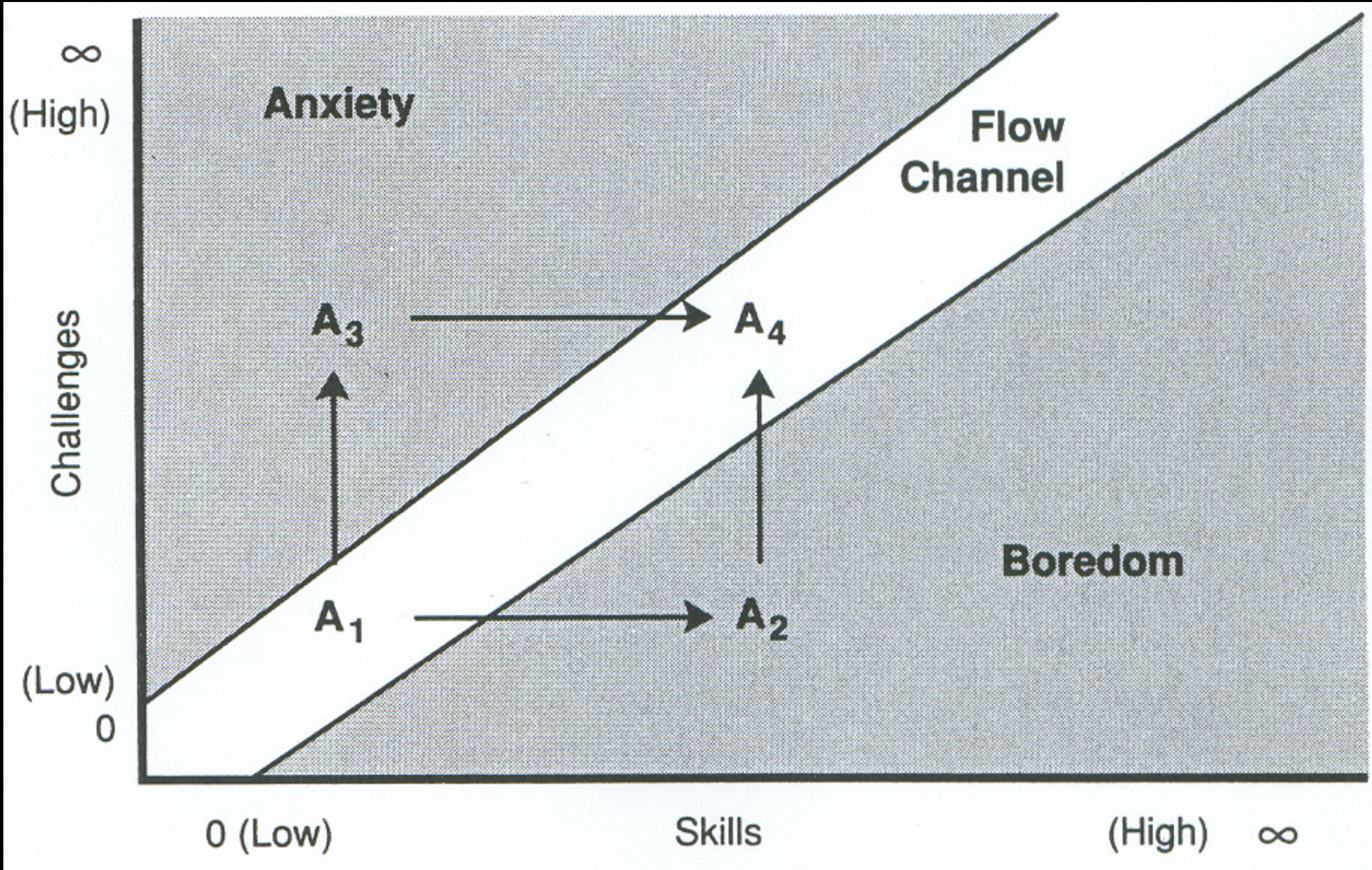
Sell - How

- Quality – Clarity – Beauty - Consistency
- Simplicity – Simple but no simpler – KISS
- Parsimony – Comprehensiveness – necessary and sufficient – Occam's Razor
- Rationale – Credibility – Logic
- Sound Theoretical Foundations
- Generalisability - Validity
- Traceability - Minimising the magic
- Scientific integrity – Utter honesty - Feynman



It is
YOUR
baby





BE IN THE FLOW CHANNEL

Hamel's Deconstruction of Apple

Apple	Other Companies
Be passionate.	Be rational.
Lead, don't follow.	Be cautious.
Aim to surprise.	Aim to satisfy.
Be unreasonable.	Be practical.
Innovate incessantly.	Innovate here and there.
Sweat the details.	Get it mostly right.
Think like an engineer, feel like an artist.	Think like an engineer, feel like an accountant.

In any endeavor you can
only be five things.

Bad

Average

Good

Excellent

Elite

Clint Bruce

ONLY DEAD FISH
GO WITH
THE FLOW

**There is no such thing
as a failed experiment,
only experiments with
unexpected outcomes.**

Buckminster Fuller

go to work, send your kids to school
Follow Fashion, act normal
walk on the pavement, watch T.V.
Save for your old age, obey the law
Repeat after me: I am free

Quality is not an act, it is a habit.

Aristotle

QUALITY

Quality means doing it right when no one is looking. *Henry Ford*

Two roads diverged in a yellow wood,
And sorry I could not travel both
And be one traveler, long I stood
And looked down one as far as I could
To where it bent in the undergrowth;
Then took the other, as just as fair
And having perhaps the better claim,
Because it was grassy and wanted
wear;

Robert Frost

Out of clutter find
simplicity. Out of discord
make harmony. Out of
difficulty find
opportunity.

Albert Einstein

**Any darn fool can make
something complex; it
takes a genius to make
something simple.**

Pete Seeger

Real artists ship!

Steve Jobs

An old saying at Apple Computer, attributed to Steve Jobs, meaning that it is important to actually deliver.

**Anybody who thinks a little
9,000-line program that's
distributed free and can be
cloned by anyone is going to
affect anything we do at
Microsoft has his head screwed
on wrong.**

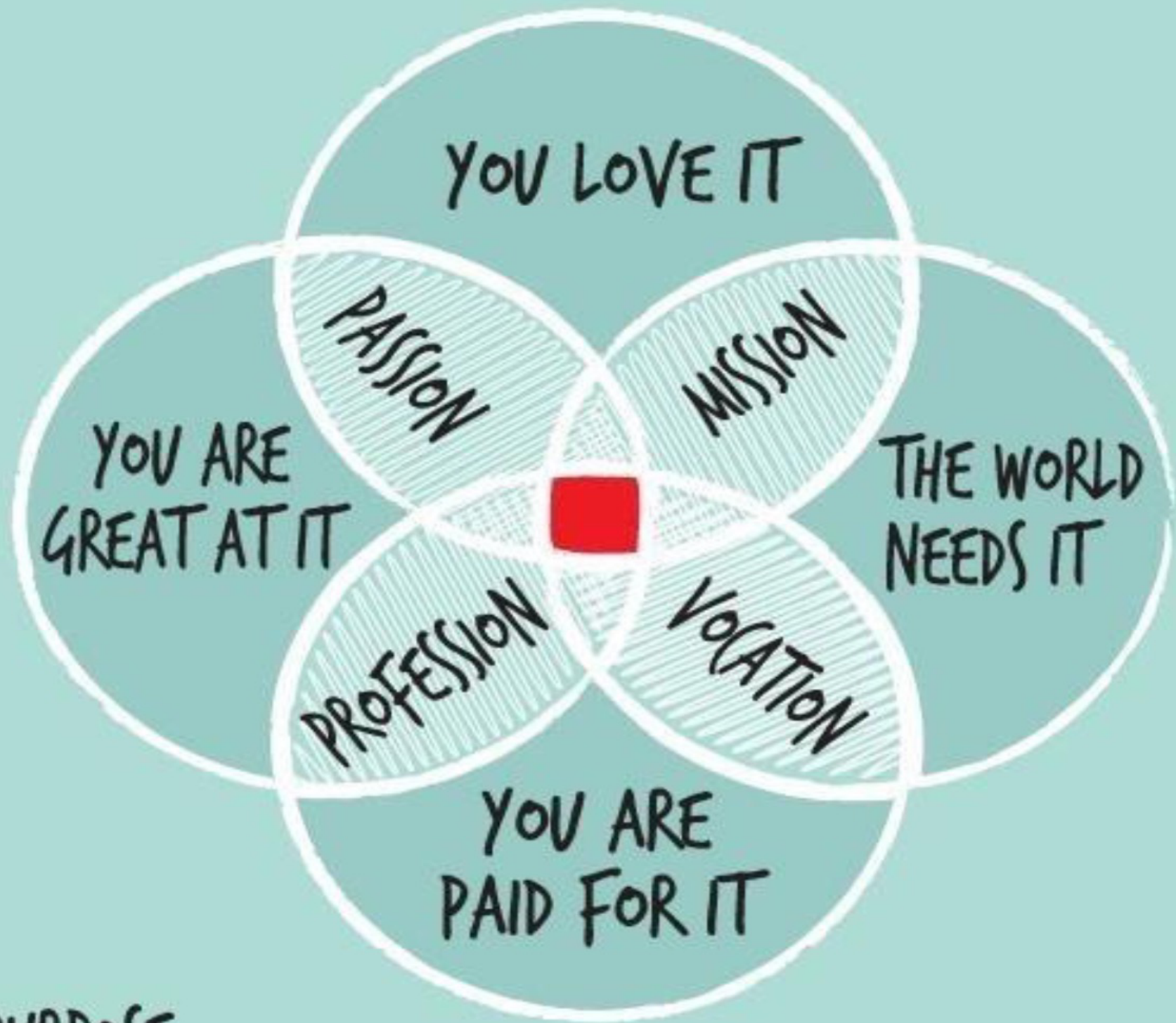
Bill Gates

**Not knowing it was
impossible, he went
ahead and did it.**

John Cocteau

**The only way of
discovering the limits of
the possible is to venture
a little way past them
into the impossible.**

Arthur C. Clarke



 -PURPOSE

**If a man does not keep pace
with his companions, perhaps it
is because he hears a different
drummer. Let him step to the
music which he hears, however
measured or far away.**

Henry David Thoreau

“The only people for me are the mad ones, the ones who are mad to live, mad to talk, mad to be saved, desirous of everything at the same time, the ones that never yawn or say a commonplace thing, but burn, burn, burn like fabulous yellow roman candles exploding like spiders across the stars.”

Jack Kerouac, *On The Road*

Don't do

IT

..if you are

NOT

passionate

about IT!

Viewed as work, research is a pretty unattractive profession. Find another view of it. I suggest you see it as play.

Play it

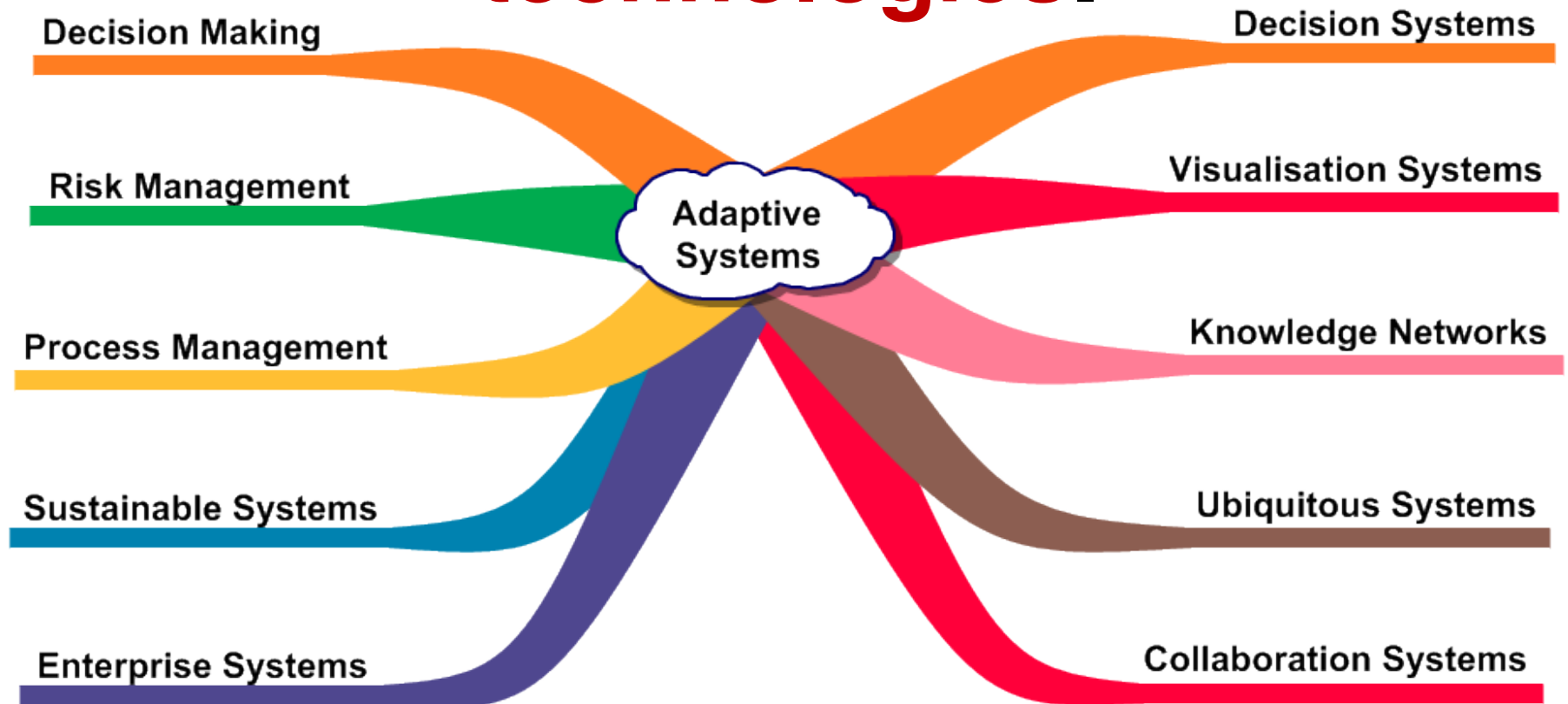
Practice it

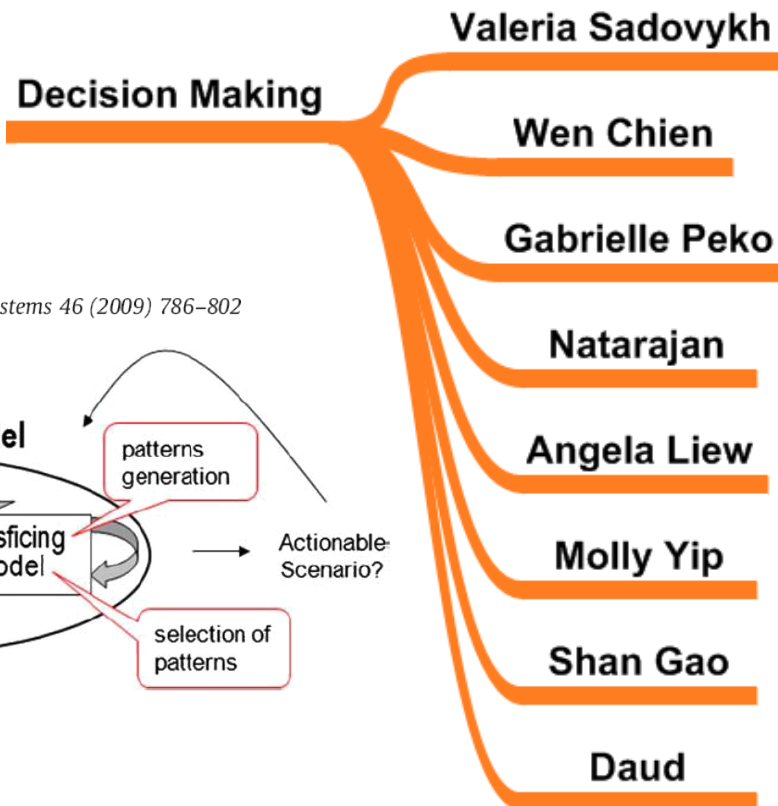
Enjoy it

Love it

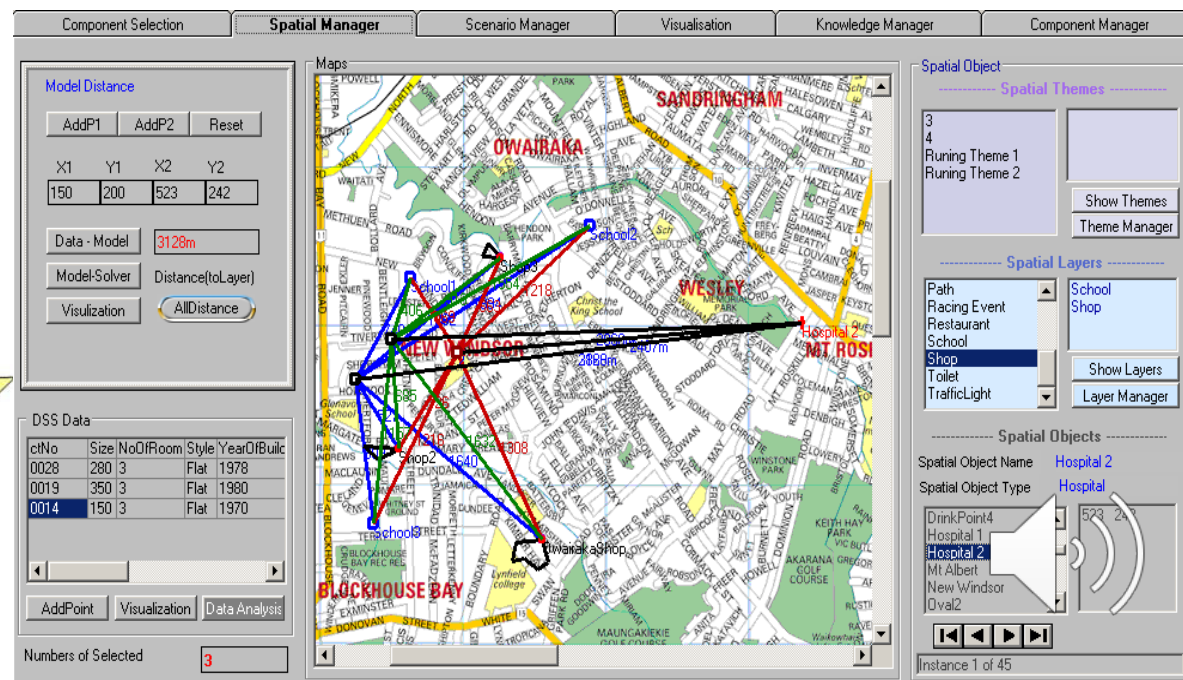
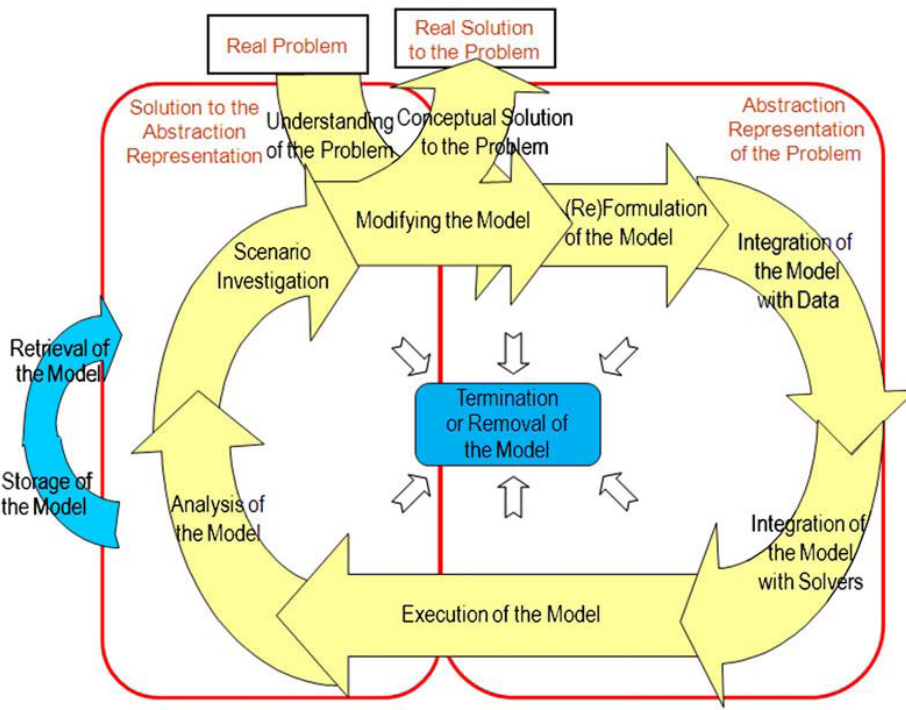
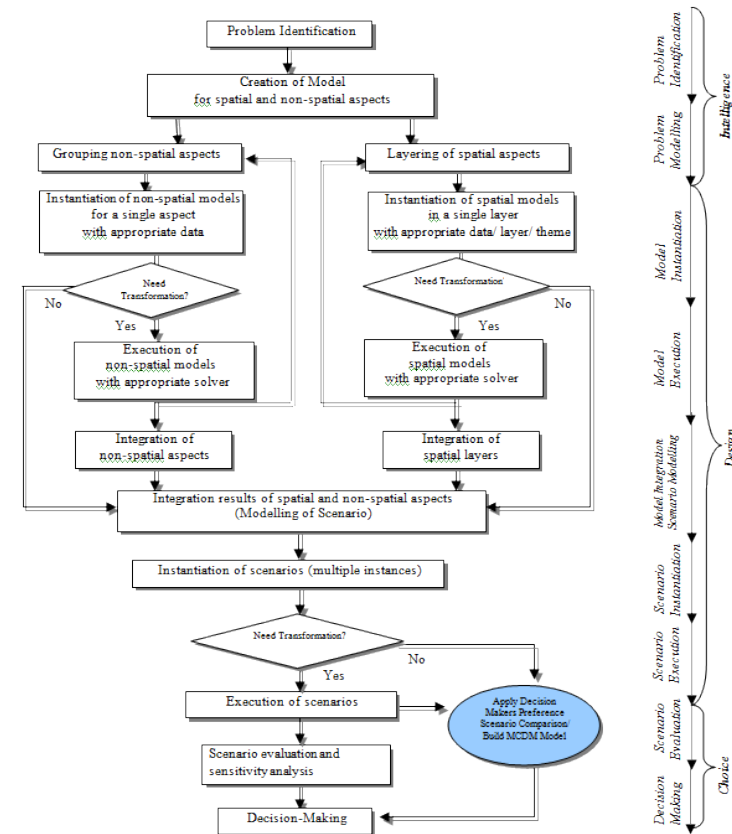
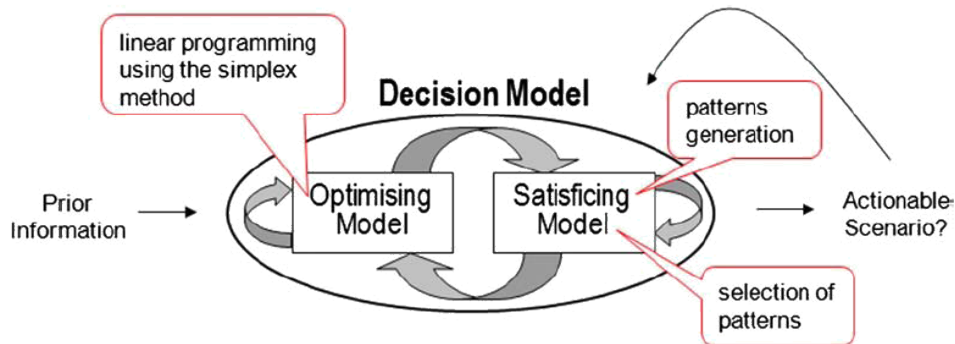
**Best Wishes on
your Design
Science Research
journey!!!!**

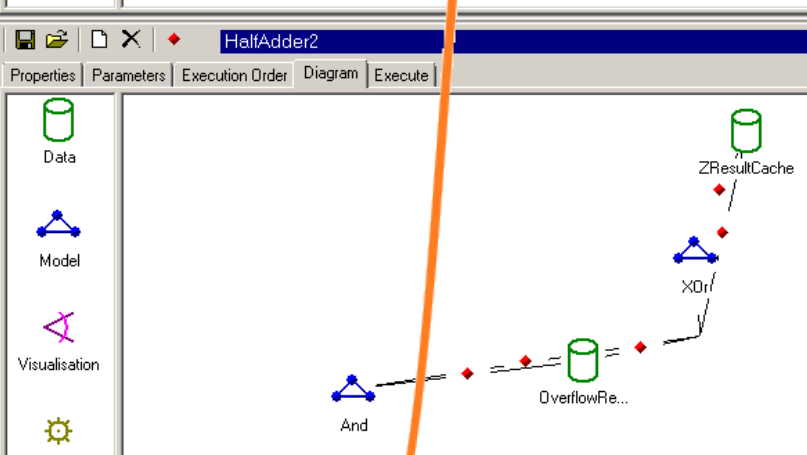
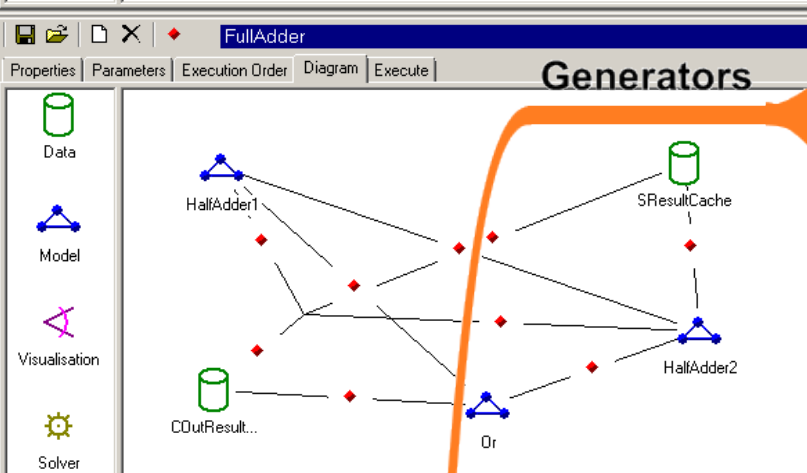
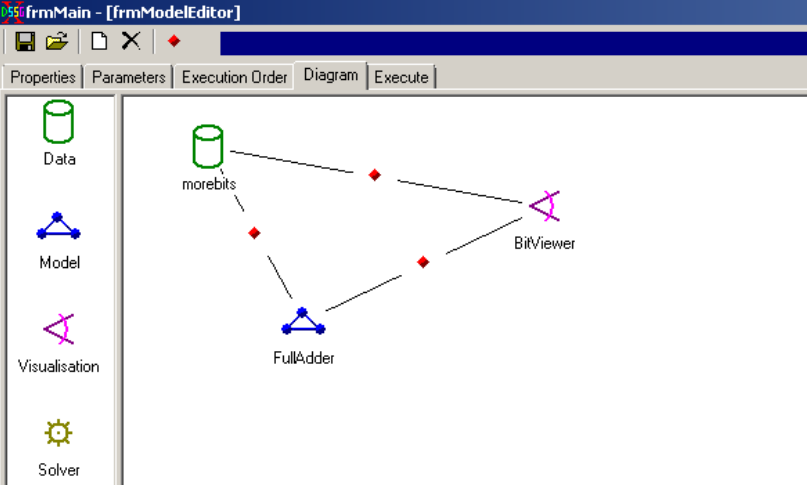
I am interested in the design and implementation of **adaptive sustainable systems** – interweaving **strategy**, **people**, **processes**, and **technologies**.





A. Liew, D. Sundaram / Decision Support Systems 46 (2009) 786–802





Decision Systems

Domain Specific

Integrated Model Management Systems

EDSSG - Allen

DSSG-FX - Stephen, Sarah, Doug

WDSSG - James

Scenario-driven - Daud

Dynamic DSSG - Dennis

Visual Modelling Environment - Doug

S-D - Molly

Spatial - Shan

AI

Agent-enabled - James

Genetics - Michael

Neural networks - Wei

Telco

Knowledge - Maron

Data-mining - Chris

Marketing - PETAPS

Cutting Stock - Liew

SARA - James

Visualisation Systems

Xiaoyan Bai

David White

Claire Donald

Michael Cooling

Problem Context

*Problem situation;
Knowledge types;
Knowledge management
tasks;
Visualization tasks;*

*Knowledge generation,
representation and presentation;
Problem-related purpose and time;
Location;
Physical surroundings;
Movement state*

Sense

Learning
Loops

Respond

Problem
Components

Visualization
Components

Sense

Presentation
Language

Learning
Loops

Respond

Action
Language

Knowledge Worker Context

*Knowledge worker type;
Knowledge worker profile;
Knowledge worker ability;*

*Knowledge worker related
purpose*

Knowledge Visualization Context

Decisional Problem

Sense

Respond

Problem Management

Problem
Data

Problem
Model

Problem
Solver

Problem
Scenario

Kernel

Visualization
Data

Visualization
Model

Visualization
Solver

Visualization
Scenario

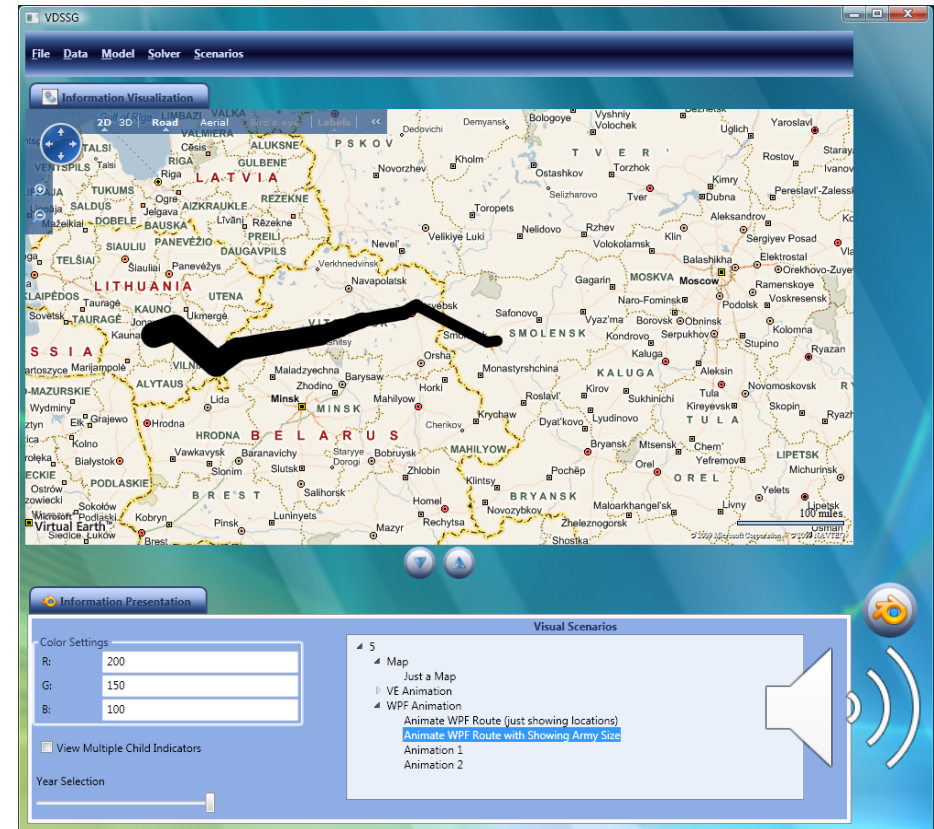
Visualization Management

User Interface

Sense

Respond

Knowledge Worker



Integrated Personal Recommender - Ronald

Collaboration Systems

Personalization

Jurgen

Conflict Resolution

Peter

Organisation Decision Making Systems

Stephen

Child Rights Education and Advocacy through Engagement

ACYA

James

Andrew

Max

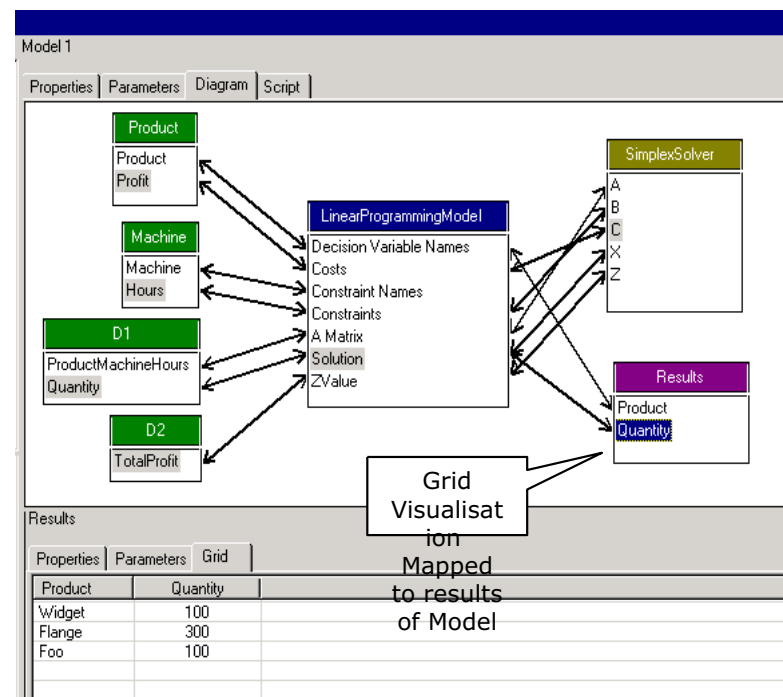
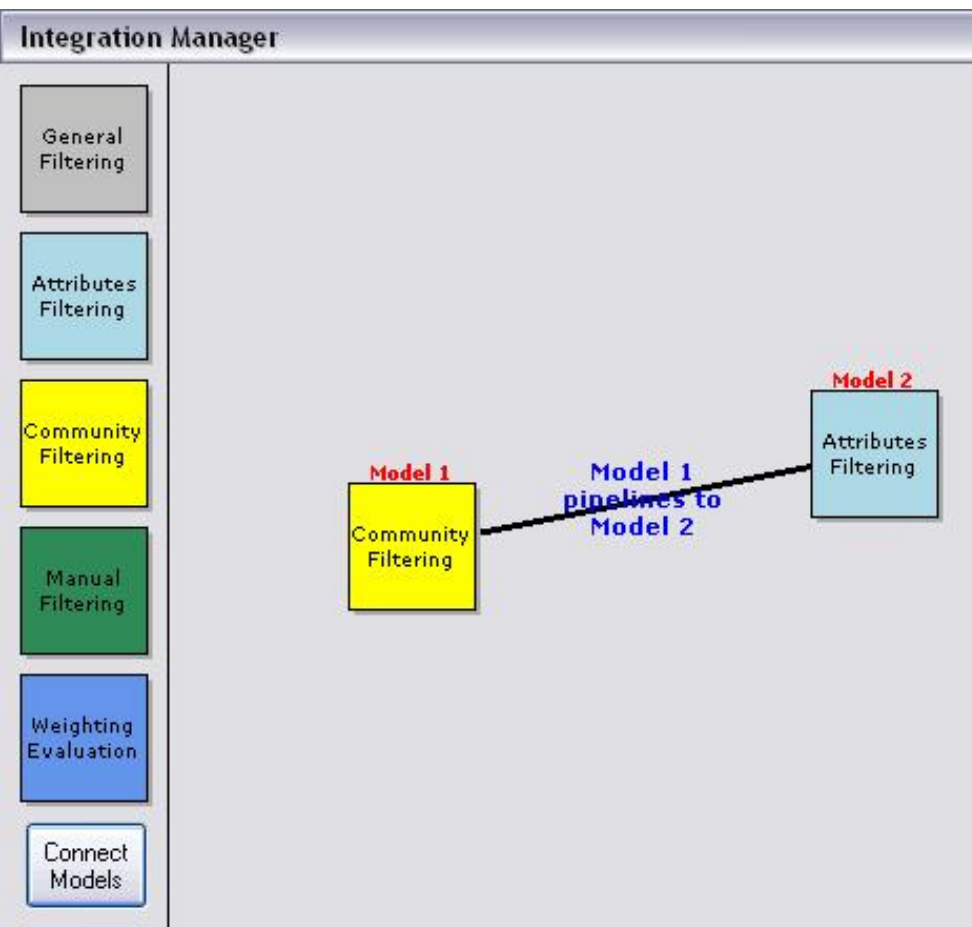
Shahper

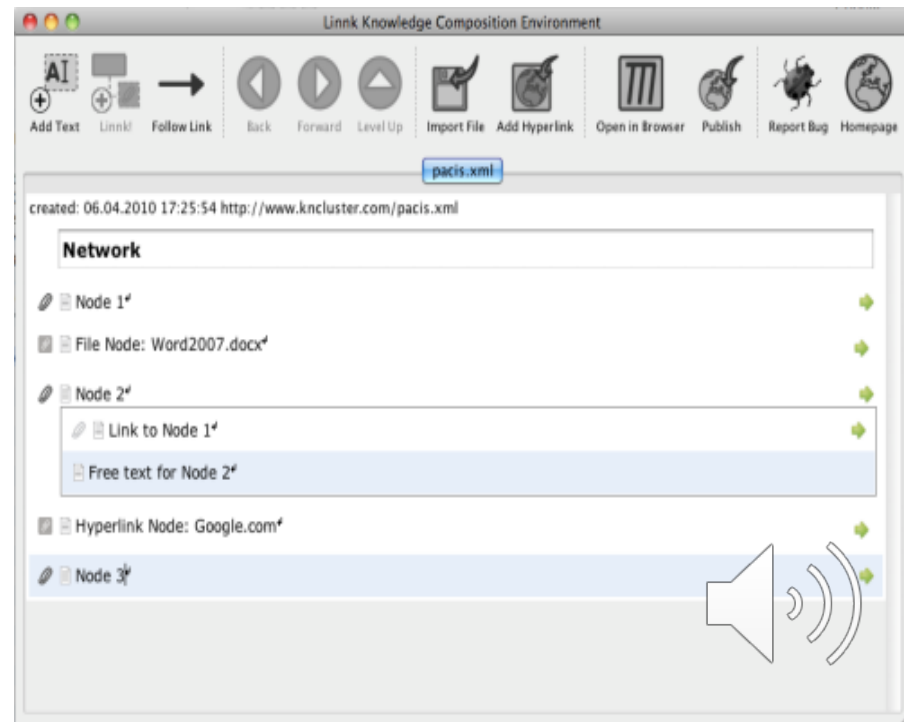
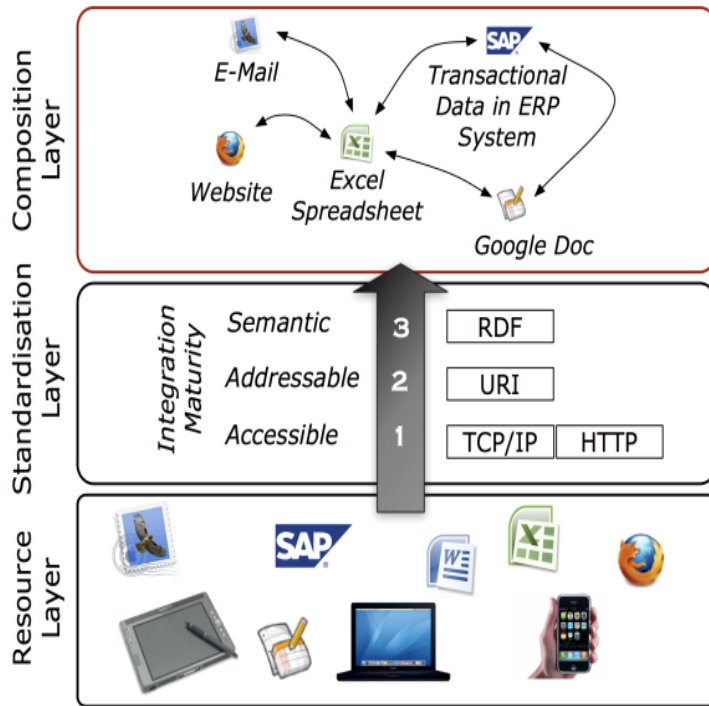
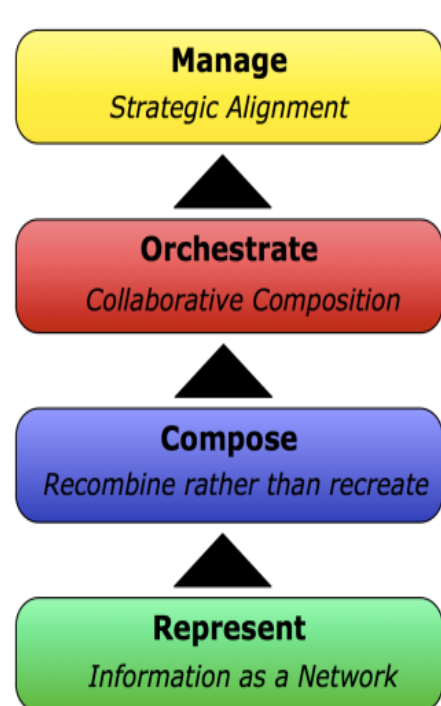
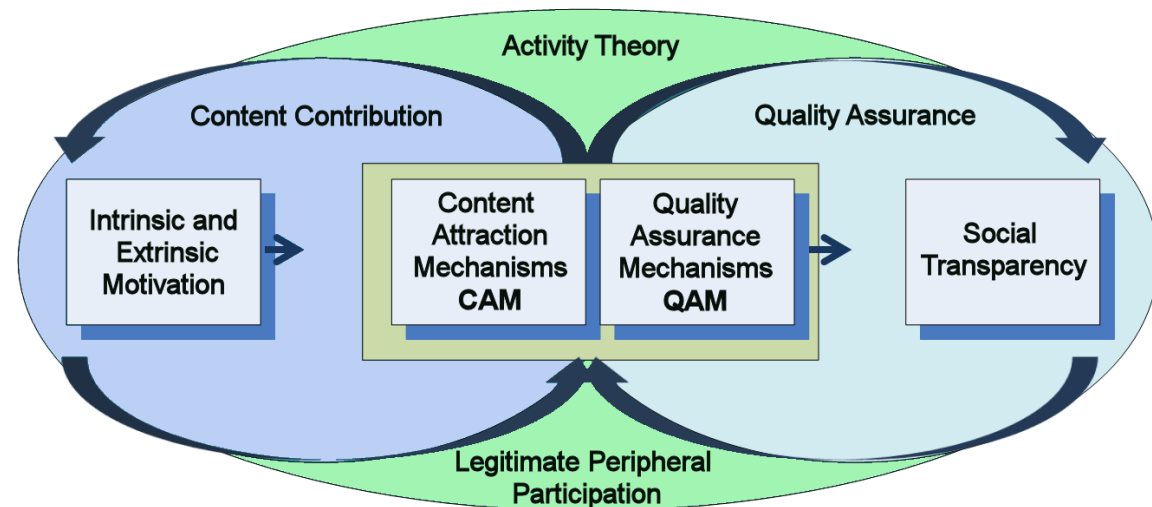
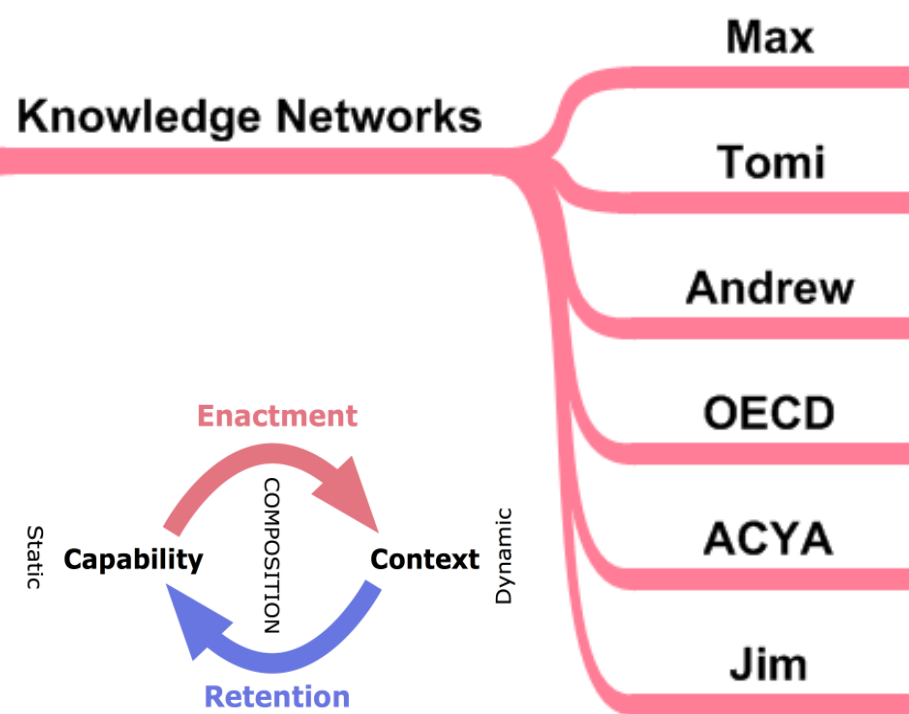
OECD

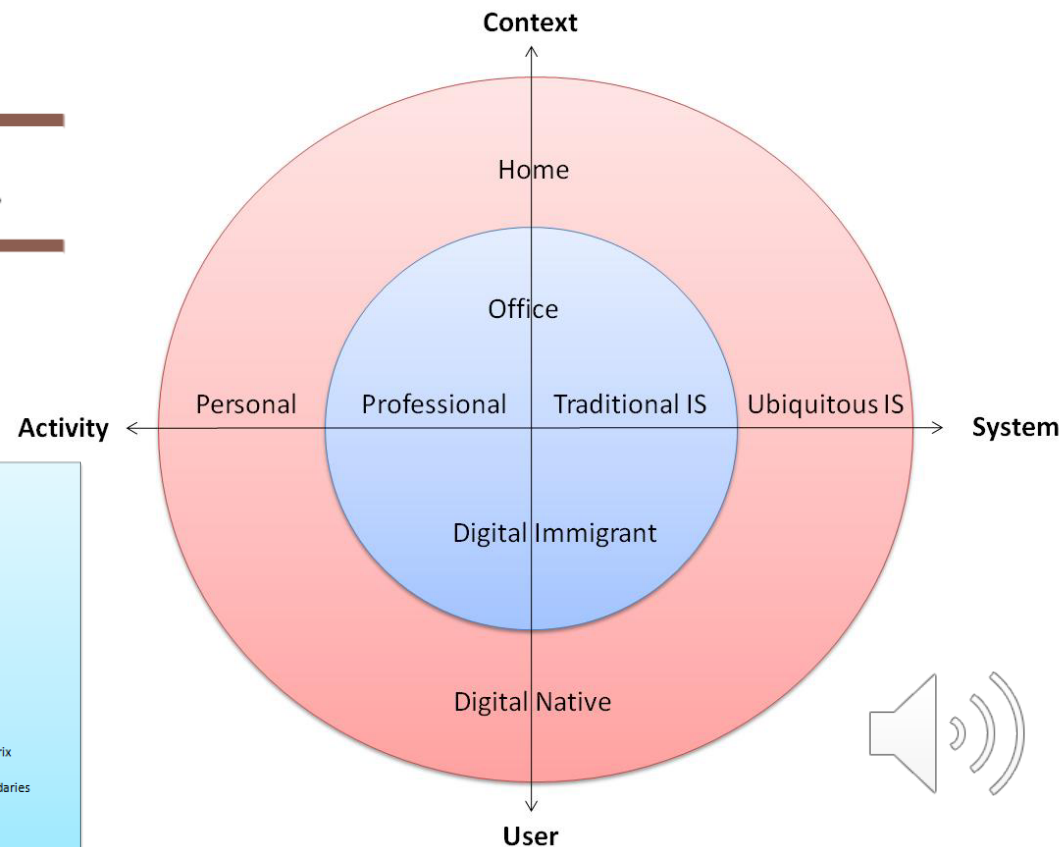
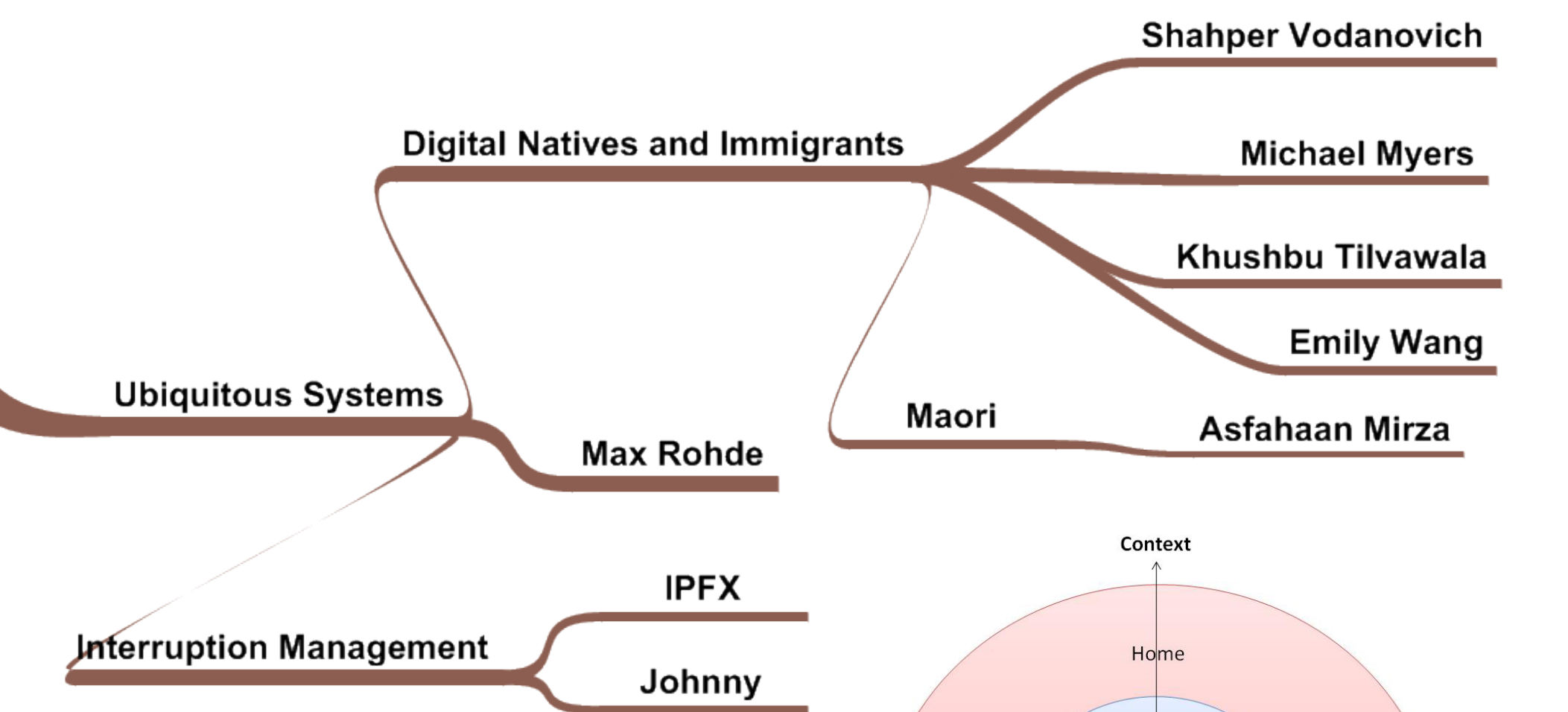
Online Content-oriented Collaborative Platforms

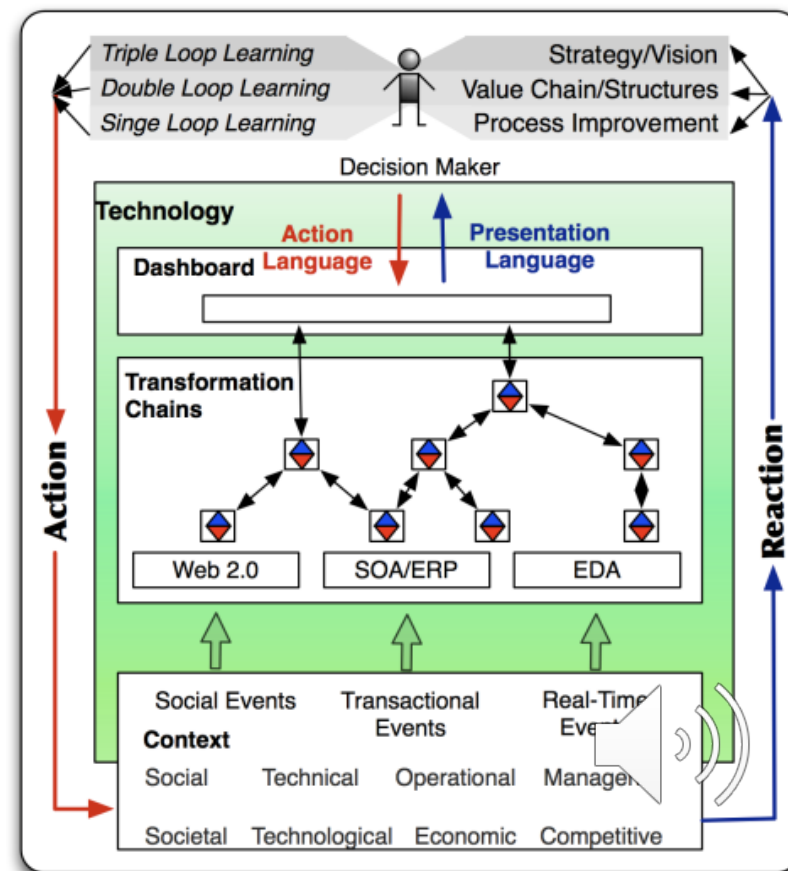
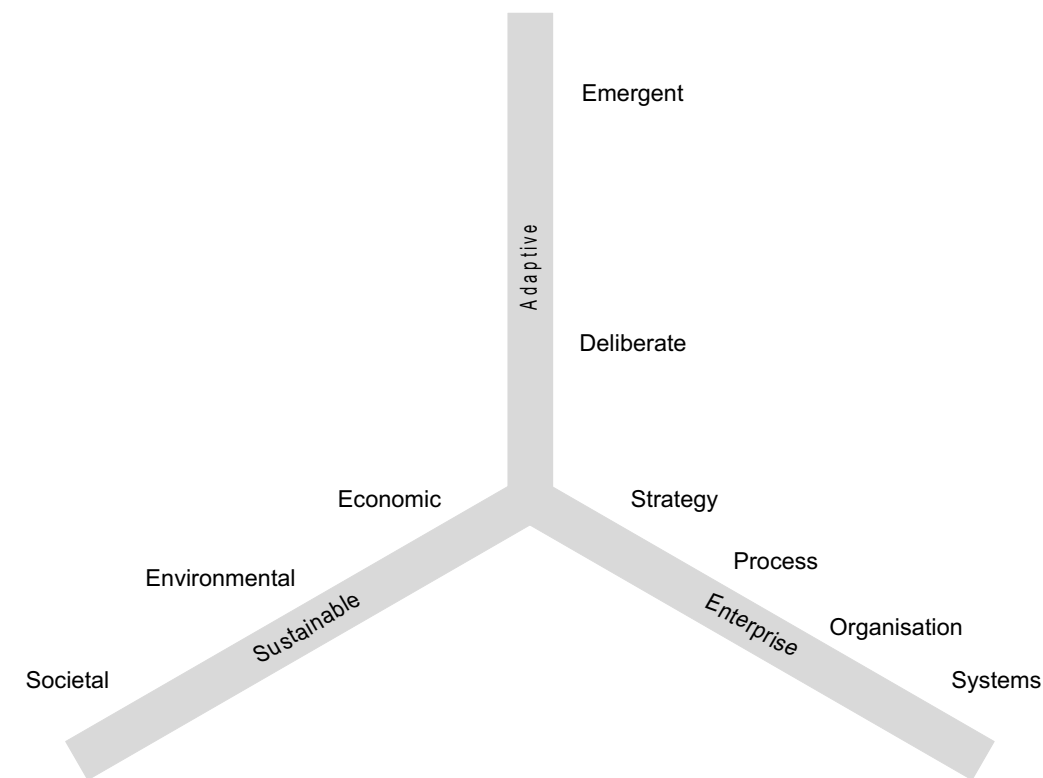
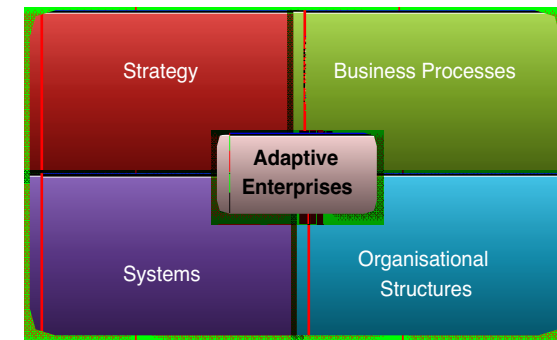
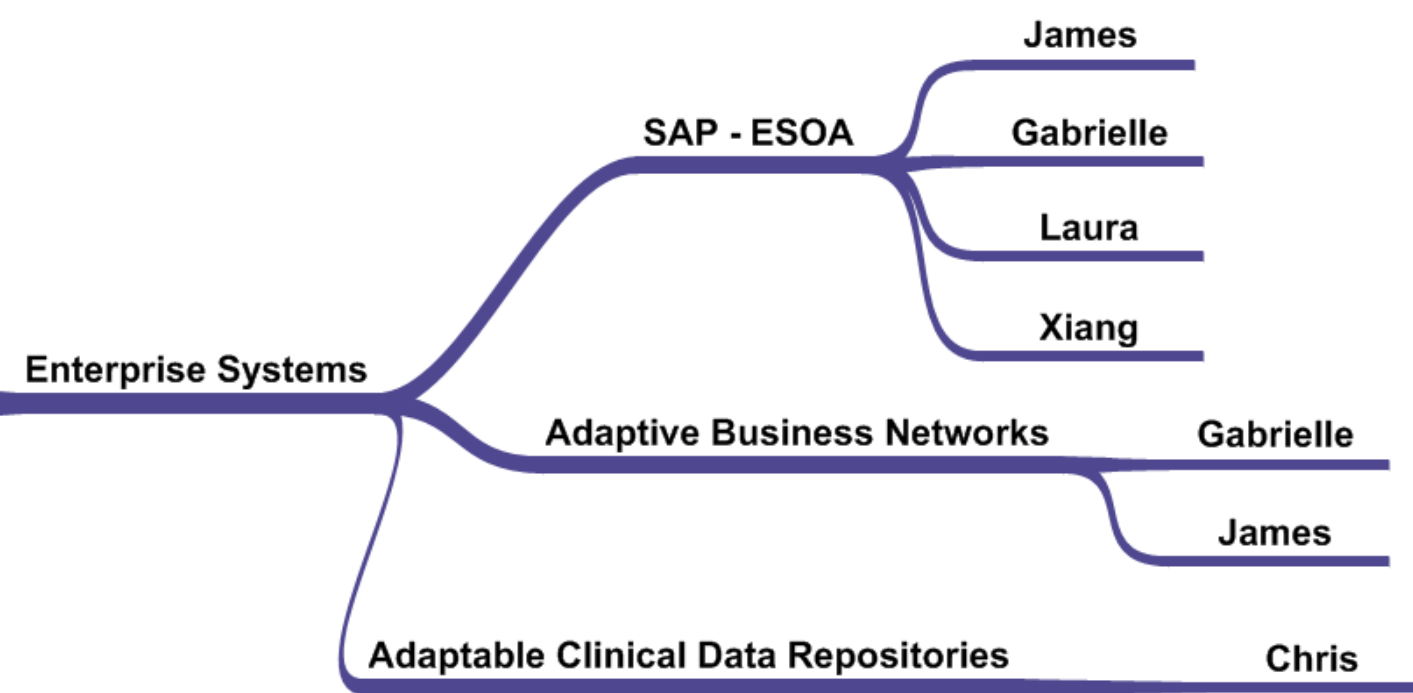
Tomi

Max









Risk Management

Operational Risk - Elke

Complexity - Martin

Agent-based- Paul

Banks - Reddy

Assessment and Management- Eric

Process Management Chasm

Known
Expected
Healthy

Unknown
Unexpected
Risky

Mal-process

Best Practice

End of Year

Prepare Performance Report

Creation of Biased Report

Employee Concerned

End of Year

Prepare Performance Report

Creation of Report

Independent Superior

Business Process Management

Risk Management

Mal-processes

Max

Victor

Eleshea

Process Management

Business Process Modelling

Max

Xiang

